A RANK-TWO ELLIPTIC CURVE OF CONDUCTOR 6503188012738584: CERTIFIED ARITHMETIC, CONDITIONAL CRITERIA, AND THE REMAINING BIRCH–SWINNERTON-DYER OBLIGATIONS

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ABSTRACT. We study the elliptic curve $E\colon y^2 = x^3 - 22735x^2 + 106272484x$ over $\mathbf{Q}$, of conductor $N = 2^3 \cdot 3 \cdot 2243 \cdot 4547^2 \cdot 5843$. We record what is proved about the Birch–Swinnerton-Dyer conjecture for E, what is proved only conditionally, and exactly what remains.

Proved.

- $E(\mathbf{Q}) \simeq \mathbf{Z}/2 \oplus \mathbf{Z}^2$ with an explicit basis, and the analytic rank is exactly 2; the analytic-rank statement is a computer-assisted certificate for $L''(E,1)$.
- The BSD quotient $q = L''(E,1)/(2\Omega_E \operatorname{Reg})$ lies in $(35.99427, 36.00573)$.
- The p-primary parts of $Ш(E/\mathbf{Q})$ for $p \le 47$ are 0 except at $p = 3$, where $Ш(E/\mathbf{Q})[3^\infty] \simeq (\mathbf{Z}/3)^2$; at $17 \le p \le 47$ the certificates apply a theorem of Ota as stated in its arXiv version.

Conditional.

- For every good prime $p \ge 5$, $p \nmid 7N$, the nonvanishing modulo p of one anticyclotomic Gross-point coefficient $M_2(p)$ implies $Ш(E/\mathbf{Q})[p^\infty] = 0$. The algebraic core of the supersingular case is checked in Lean.
- At the split multiplicative prime 2243, a single coefficient, not yet computed, would prove $Ш(E/\mathbf{Q})[2243^\infty] = 0$ if it is nonzero modulo 2243; a zero value would prove nothing. Every stated theoretical input is supplied at its recorded import or certificate grade, including an independent certificate that the split 2243-adic height is nondegenerate.

Open.

- Full BSD for E remains open: it requires Ш finite of order 9 and the exact identity $q = 36$. Finiteness is also tracked as a separate intermediate milestone.

We also record a family of scoped obstructions, and separate new results from applications of published theorems. Each principal claim carries an identifier into the accompanying claim-to-evidence index.

Private evidence release. `https://github.com/Bome2017/bsd-seed-research`. Tag: `v0.2.2-private-review`. This is a provisional research dossier for targeted expert review, not an independently refereed proof paper. Repository access requires authorization. The quick replay verifies included downstream evidence; the large numerical corpus is external and is not reevaluated by that command.

Local revision v0.2.5 (unreleased, 29 September 2026) of the private-review release above. Relative to v0.2.4 it adds the singleton certificates at $17 \le p \le 47$ (Theorem 4.15); finiteness of $Ш[3^\infty]$ and the lower bound $Ш[3] \supseteq (\mathbf{Z}/3)^2$ without the main conjecture (Theorems 4.6 and 4.7); an exact statement of the provenance at 3 (Remark 4.9); route K, a proof of $\#Ш[3^\infty] \le 9$ without the main conjecture that becomes the primary support of Theorem 4.5 (Theorem 4.8); the bad-prime Criterion 8.7; and Research records 9.7, 9.8 and 9.9. Revision v0.2.4 (25 September 2026) added Remarks 4.4 and 4.10 (formal checks of the deductions at $p = 3, 5, 7, 11, 13$), an exact descent certificate for the rank and $Ш[2]$ (Theorem 3.1), and the corresponding claim grades. The date and tag above are those of the base release v0.2.2; this revision is a local working copy, not yet published, with no tag or snapshot link. Base mathematical source snapshot: `bf75979a9fb1b53af7868c8c1a31eb5bc010e8ab`.

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1. Introduction

1.1. **The curve.** Let

$$E\colon y^2 = x^3 - 22735x^2 + 106272484x, \qquad N = 2^3 \cdot 3 \cdot 2243 \cdot 4547^2 \cdot 5843 = 6503188012738584.$$

The curve arises in a family studied by Zywina [43]. It has rational 2-torsion, root number $+1$, and a Mordell–Weil group of rank two with an explicit basis. Its Birch–Swinnerton-Dyer quotient is certified to lie within 0.006 of 36.

This paper does three things for E:

- it collects what is proved about the Birch–Swinnerton-Dyer (BSD) conjecture;
- it states the conditional criteria that have been established;
- it states precisely what remains open.

We call E a *candidate seed* only informally. No statement below should be read as more than its stated grade.

1.2. **What is proved.** Write $Ш = Ш(E/\mathbf{Q})$, Reg for the Néron–Tate regulator of the basis in Theorem 3.1, $\Omega_E = 2\omega_1$ for the full real period, and

$$q = \frac{L''(E,1)}{2\,\Omega_E\,\mathrm{Reg}}.$$

Theorem 1.1 (Rank and the BSD quotient)**.** *(i)* $E(\mathbf{Q}) = \langle (0,0) \rangle \oplus \mathbf{Z}P_1 \oplus \mathbf{Z}P_2$ *with* $P_1 = (4547, 327384)$ *and* $P_2 = (5184, 281520)$.
(ii) $\mathrm{ord}_{s=1} L(E,s) = 2 = \mathrm{rank}\, E(\mathbf{Q})$.
(iii) $35.99427 < q < 36.00573$.
(iv) Full BSD for E *holds if and only if* $Ш$ *is finite of order* 9 *and* $q = 36$.

Part (i) combines an exact 2-descent, computed by two independent implementations, with the separate global saturation certificate and its replay. Zywina's family theorem [43] corroborates it. Part (ii) combines a rigorous interval certificate for $L''(E,1)$ with standard results. Part (iii) combines that certificate with a certified regulator. Part (iv) is elementary. See Theorems 3.1–3.7.

Theorem 1.2 (Primary components)**.** $Ш[p^\infty] = 0$ *for every prime* $p \le 47$ *with* $p \neq 3$*, and* $Ш[3^\infty] \simeq (\mathbf{Z}/3)^2$.

These computer-assisted claims retain the explicit symbol-lattice proof, coefficient-verifier and custody qualifications in Appendices D and F; the 13-primary status-file discrepancy is not suppressed. The components at $3, 5, 7, 11, 13$ come from certified depth-two (at 3, depth-four) Mazur–Tate coefficients. They combine with Kato's divisibility at the good ordinary primes. At 3 they combine with Kato's Euler system and an integral Coleman map (route K), and, as a second route, with Skinner's multiplicative main-conjecture theorem. The components at $17 \le p \le 47$ come from certified singleton Mazur–Tate coefficients, one auxiliary prime each, through a theorem of Ota [32] applied as stated in its arXiv version. See §4. At 3 the order 9 is proved without any main conjecture, by route K (Theorem 4.8, a written derivation read

by three Anthropic model readers and in one cross-family source reading, but not by a human referee) together with explicit 3-coverings. Skinner's Theorem A, as printed, gives a second route, two steps of whose printed proof are not established for this curve (Remark 4.9).

1.3. **What is proved conditionally.**

Theorem 1.3 (Forward bridges; conditional)**.** *For every prime $p \geq 5$ of good reduction with $p \nmid 7N$ there is an explicit anticyclotomic quantity $M_2(p) \in \mathbf{F}_p$, built from Gross points of conductor p^2 over $\mathbf{Q}(\sqrt{-7})$, such that*

$$M_2(p) \neq 0 \quad \Longrightarrow \quad Ш[p^\infty] = 0.$$

The supersingular case has a written proof whose algebraic core is checked in Lean 4. The ordinary case has two written proofs. No prime is settled by Theorem 1.3: nonvanishing of $M_2(p)$ is open at every prime, and we know of no method to evaluate it on this curve (Remark 7.7).

Theorem 1.4 (The split multiplicative prime 2243; conditional)**.** *There is an explicit integer K, a weighted sum of plus modular symbols at the cusps $b/2243^2$, such that $K \not\equiv 0 \pmod{2243}$ implies $Ш[2243^\infty] = 0$. Its theoretical inputs are supplied at the source-import and computation grades listed in §8. The coefficient has not been computed; its cost is about 10^5 core-hours. A zero value would prove nothing.*

1.4. **What remains.** Full BSD for E is equivalent to three open statements (Theorem 1.1(iv)):

(a) **FIN**: Ш is finite;
(b) **ORDER9**: its order is 9;
(c) **Q36**: $q = 36$.

Given FIN, the settled components force $\#Ш = 9t^2$ with every prime factor of t at least 53, so the next possible order is $25281 = 9 \cdot 53^2$. The unsettled primes fall into three regimes:

- the good ordinary primes $p \geq 53$;
- the good supersingular primes $p \geq 53$ $(131, 167, 281, \dots)$, of which there are infinitely many;
- the three bad primes 2243, 4547, 5843.

Section 11 states these obligations with quantifiers and describes the structure of the problem as the whole record shows it.

1.5. **How to read the grades.** The principal claims carry identifiers in typewriter type, for example `R.3`. The identifier points to the entry in `claims/CLAIM_INDEX.md` of the accompanying repository, which lists:

- the exact supporting artifacts, with SHA-256 hashes;
- the imported theorems, and how closely they were checked at their sources;
- the normalization;
- what has been replayed or reviewed;
- every qualification.

We use the following grades, and never merge them:

- *published theorem, applied*: we apply a published result; the index records whether its statement was checked at the source.
- *preprint theorem, applied*: the same, for an unrefereed preprint.
- *written proof*: an argument given here or in the cited records, possibly combining imported results.
- *computer-assisted*: a certificate with rigorous error control (interval arithmetic or exact arithmetic), with a stated trust base.

- *Lean-checked*: a deduction checked by Lean 4 from hypotheses that are listed and are not themselves formalized.
- *conditional*: an implication whose antecedent is not established. Such an antecedent is a *sufficient witness*, never an obligation: its failure proves nothing.

No statement in this paper has yet received independent human review. Several statements were reviewed by automated (language-model) reviewers, in some cases of a different model family from the author's tools; the index records which. Stopped research routes are graded separately (Appendix A). A *proved obstruction* is a theorem about an explicitly delimited class; an *unaffordable* route is a budget decision; an *unsuccessful attempt* claims nothing.

1.6. **What is new.** The release separates four kinds of contribution. First, the rank and fixed-prime conclusions apply existing descent and Iwasawa methods to this high-conductor curve; rank-two primary computations are not a new methodology by themselves. Second, the numerical contribution is the curve-specific certificate implementation and its explicit, reviewable trust chain. Third, the equivariant deduction in Appendix E is proposed mathematical work built on BLV's arithmetic inputs; its novelty has not been independently established. Fourth, the archived equivalences and stopped constructions document limits of the methods examined and are not evidence that every route in those broad subjects fails.

The curve offers an explicit rank-two basis, nontrivial 3-primary Sha, both multiplicative types and an additive potentially supersingular prime. These make its comparison of local regimes concrete. Its high conductor is also a serious computational disadvantage, not evidence by itself that mass computation will close BSD. Whether these features make it a useful research seed is left for mathematical assessment.

1.7. **Organization.** §2 fixes the curve. §3 proves Theorem 1.1. §4 proves Theorem 1.2. §5 records the exact endgame for q. §6 certifies two quadratic twists. §§7–8 give the conditional criteria. §9 records equivalences. §10 records obstructions. §11 states what remains. The appendices describe the grades, the computations and their trust base, the full lattice and A3 arguments and the computational custody chain. Historical source sections and unrelated-curve records remain in the repository, outside the main PDF.

2. The curve

Proposition 2.1 (`A.1`; exact computation)**.** *The following data hold for* $E\colon y^2 = x^3 - 22735x^2 + 106272484x$.

(i) Minimal model. *The substitution* $x = X + 7578$ *(with* $u = 1$*, so the invariant differential is unchanged) gives the global minimal model*

$$y^2 = X^3 - X^2 - 66020924X - 65074271436.$$

Since $X = x - 7578$*, the basis points* $P_1 = (4547, 327384)$ *and* $P_2 = (5184, 281520)$ *of Theorem 3.1 have coordinates* $(-3031, 327384)$ *and* $(-2394, 281520)$ *on this model. Its invariants are* $c_4 = 3169004368$*,* $c_6 = 56243184546880$ *and*

$$\Delta = 2^8 \cdot 3^2 \cdot 2243 \cdot 4547^3 \cdot 5843^2 = 16586638656786071143743744 > 0.$$

(ii) Bad reduction. *The bad primes behave as follows.*

ℓ	*Kodaira type*	c_ℓ	a_ℓ	*remark*
2	I_1^*	2	0	*additive*
3	I_2	2	-1	*nonsplit multiplicative*
2243	I_1	1	$+1$	*split multiplicative*
4547	III	2	0	*additive, potentially good supersingular*
5843	I_2	2	-1	*nonsplit multiplicative*

The Tamagawa product is 16. *At* 4547 *one has* $j \equiv 1728$, $4547 \equiv 3 \pmod 4$, *and inertia acts through a cyclic group of order* 4.

(iii) Global invariants. $E(\mathbf{Q})_{\mathrm{tors}} = \mathbf{Z}/2$, *generated by* $T = (0, 0)$. *The root number is* $+1$.

(iv) Good primes. *At every good prime* p *the trace* a_p *is even, and* $p \nmid \#E(\mathbf{F}_p)$. *The good supersingular primes below* 2000 *are*

$$23,\ 29,\ 131,\ 167,\ 281,\ 563,\ 661,\ 823,\ 839,\ 863,\ 887,\ 1013,\ 1471,\ 1511,\ 1823.$$

Justification. Items (i) and (iv) (for $p < 2000$) are recomputed in exact integer arithmetic by `tools/replay/curve_invariants.py`, which uses no computer algebra system. The Kodaira types, Tamagawa numbers and root number are the output of Tate's algorithm and the local root-number formulas, recorded in `certificate/hypotheses_output.txt`.

For (iv) in general: the rational 2-torsion point makes $\#E(\mathbf{F}_p)$ even, so $a_p = p + 1 - \#E(\mathbf{F}_p)$ is even. If $p \mid \#E(\mathbf{F}_p)$, then $\#E(\mathbf{F}_p)$ is a positive even multiple of p in $[p+1-2\sqrt{p},\ p+1+2\sqrt{p}]$. That forces $2p \le p + 1 + 2\sqrt{p}$, hence $p \le 5$; and $\#E(\mathbf{F}_5) = 8$. □

Proposition 2.2 (`A.2`; exact computation)**.** $j(E) = 1322372123614064/689197281363 \notin \mathbf{Z}$, *so* E *has no complex multiplication.*

Theorem 2.3 (`A.3`; written proof applying published theorems)**.** *For every odd prime* ℓ, *the mod-*ℓ *representation* $\bar\rho_{E,\ell}\colon \mathrm{Gal}(\bar{\mathbf{Q}}/\mathbf{Q}) \to \mathrm{GL}_2(\mathbf{F}_\ell)$ *is surjective. Hence the* ℓ*-adic representation is surjective onto* $\mathrm{GL}_2(\mathbf{Z}_\ell)$ *for* $\ell \ge 5$.

Proof. Reducibility would give a rational cyclic isogeny of odd prime degree ℓ. The prime-isogeny theorem restricts ℓ to $3, 5, 7, 11, 13, 17, 19, 37, 43, 67, 163$. Direct point counts give the following witnesses:

ℓ	q	a_q	$a_q^2 - 4q$
$3, 7, 11, 19, 43, 67, 163$	5	-2	-16
$13, 17, 37$	7	-2	-24
5	17	-6	-32

Every displayed discriminant is a nonsquare modulo each listed ℓ, and $q \ne \ell$. Thus the indicated Frobenius has no invariant line, contradicting reducibility. These finite checks are reproduced in `review/pro_report/independent_checks.py`.

For $\ell \ne 2243$, multiplicative reduction with $v_{2243}(\Delta) = 1$ supplies a nontrivial Tate transvection. For $\ell = 2243$, use instead 5843, with $v_{5843}(\Delta) = 2$ prime to ℓ; nonsplit multiplicative reduction is split over an unramified quadratic extension, so its inertia gives the same conclusion. An irreducible subgroup containing a transvection also contains a conjugate fixing a different line. In a basis of those two lines their powers give all upper and lower elementary unipotents over the prime field $\mathbf{F}_\ell$. These generate $\mathrm{SL}_2(\mathbf{F}_\ell)$. The determinant is the surjective cyclotomic character, proving the mod-ℓ assertion. The standard integral lifting theorem for closed subgroups at $\ell \ge 5$, with the cyclotomic determinant, gives the ℓ-adic assertion. These are the retained rational-isogeny and lifting imports of `A.3`; no lift is asserted at $\ell = 3$. □

Theorem 2.4 (A.4; written proof applying published theorems)**.** *The optimal curve in the isogeny class of E has Manin constant $c_0 = 2^a$ with $a \le 8$; in particular $4547 \nmid c_0$. The chosen parametrization ϕ_E specified in Appendix D has Manin constant c_E dividing 2^9. Consequently, with*

$$\lambda(r) = 2\pi i \int_{i\infty}^{r} f(z)\,dz, \qquad [r]^+ = \frac{\operatorname{Re}\lambda(r)}{\Omega^+}, \qquad [r]^- = \frac{\operatorname{Im}\lambda(r)}{\Omega^-} \qquad (r = a/d \in \mathbf{Q},\ (a,d) = 1),$$

where $\Omega^+ = \omega_1$ and Ω^- are the least positive real and imaginary periods, $[r]^\pm \in \frac{1}{1024}\mathbf{Z}$ whenever $\gcd(d, N)$ is an exact divisor of N. These are the rational unitary cusps needed by the retained certificates, including denominators p^n for $p \nmid N$ or $p \parallel N$.

The complete derived argument, with its geometric imports and exact chosen maps, is in Appendix D. The bound $a \le 8$ is a derived lemma; no single source states it. It combines the method of Pasten [33, §10] with [11, 6.12(c)] and [2, 3.6(a)]. The exclusion $4547 \nmid c_0$ also follows from Edixhoven's thesis [14, Thm. 4.6.3], whose statement was checked directly in the follow-up review; the published version [13, Thm. 3] was read through restatements [11]. The stronger bound $a \le 4$ would need an unverified contact-order input and is not used.

For these cusps, Manin–Drinfeld and rationality put their images in $E(\mathbf{Q})_{\mathrm{tors}}$, of exponent 2. Therefore $2c_E\lambda(r)$ is a period. The rectangular period lattice ($\Delta > 0$) gives the stated bound separately for the real and imaginary parts. This argument makes no assertion about all nonrational cusps.

Remark 2.5 (Why the lattice matters)*.* The identification step needs both a proved lattice and an interval containing exactly one point of that lattice. Radius $< 1/2048$ supplies uniqueness, not membership. An incorrect denominator, whether a power of 2 or an odd factor, can silently produce a wrong rounded value. For example an interval centered at 0 of radius 2^{-19} contains the true value 2^{-20} but has 0 as its only point of $\frac{1}{1024}\mathbf{Z}$. Thus the Manin-constant and cusp arguments remain load-bearing even when all rounding checks pass.

3. Rank and the analytic side

Theorem 3.1 (R.1; exact computation (two independent descents); global saturation computer-assisted)**.** $E(\mathbf{Q}) = \langle T\rangle \oplus \mathbf{Z}P_1 \oplus \mathbf{Z}P_2$*, with $T = (0,0)$, $P_1 = (4547, 327384)$ and $P_2 = (5184, 281520)$. The points P_1, P_2 are a basis of $E(\mathbf{Q})/E(\mathbf{Q})_{\mathrm{tors}}$, not merely generators of a subgroup of finite index. Moreover* $Ш[2] = 0$*.*

The specialization is $m = 2243$, $n = 12$: $m + 16n^2 = 4547$ and $m + 25n^2 = 5843$, all three primes being 11 modulo 24. An exact 2-isogeny descent supplies the rank and 2-descent input. Write $E'\colon y^2 = x(x^2 + 45470x + 91790289)$ for the 2-isogenous curve, and α, α' for the descent maps (x-coordinate modulo squares). Since $106272484 = 2^2 \cdot 4547 \cdot 5843$ and $22735^2 - 4 \cdot 106272484 = 3^2 \cdot 2243 \cdot 4547$, both descents are local at $\{2, 3, 2243, 4547, 5843, \infty\}$ and involve no class groups. The torsion points, P_1 and one point of $E'(\mathbf{Q})$ give $\#\alpha(E(\mathbf{Q})) \cdot \#\alpha'(E'(\mathbf{Q})) = 16 = 2^{r+2}$. This equals the Selmer bound, computed independently by PARI's unconditional `ellrank` and by eclib's `mwrank`. Hence $\operatorname{rank} E(\mathbf{Q}) = 2$ and $Ш[2] = 0$, the 2-Selmer rank being $3 = 2 + \dim_{\mathbf{F}_2} E(\mathbf{Q})[2]$. The family theorem [43, Thm. 1.2] gives the same conclusions for the whole family and is retained as corroboration. It does not itself assert that the displayed points form an integral basis. The separate Lane 3 certificate uses eclib saturation with a computed global index bound: that bound is 2, the points are proved 2-saturated, and the index is 1. A basis returned independently is P_2, G, where $G = (5843, 210348) = -P_1 - P_2 + T$; the change-of-basis determinant is 1. This calculation was replayed during the first release review. Its scripts, outputs and the corrected alternative Silverman/ratpoints proof are now included in the evidence. A bounded PARI saturation check up to 50 or 100 alone would not prove full

saturation. No preprint input remains in this theorem. The descent and the basis computation each have their own computer-assisted trust base, and the descent certificate replays in seconds (`evidence/src/BSD_PERIMETER_Q36_2026-09-25/descent/`).

Proposition 3.2 (`R.2`; published theorems applied)**.** $L(E,1) = L'(E,1) = 0$.

Proof. If $L(E,1) \neq 0$, then $E(\mathbf{Q})$ would be finite by Kolyvagin [28] (or Kato [22]); but P_1 has infinite order. The root number is $+1$, so $\operatorname{ord}_{s=1} L(E,s)$ is even and at least 2. □

Theorem 3.3 (`R.3`; computer-assisted)**.** $L''(E,1)/2 \in [35.929604,\ 35.941023]$. *Consequently* $\operatorname{ord}_{s=1} L(E,s) = 2$.

The value comes from the standard exponentially convergent series for $L''(E,1)$ [12, §2.13]. The partial sum over $n \leq 1.5 \cdot 10^8$ is enclosed in an Arb ball [19] of radius $2.65 \cdot 10^{-26}$; an elementary tail bound of $5.70901 \cdot 10^{-3}$ is then added. An outward-rounded enclosure is

$$35.93531375275170089 \pm 5.70901 \cdot 10^{-3}.$$

Its computational inputs are:

- the coefficients a_n, generated multiplicatively from 8,444,396 prime coefficients a_q;
- the check of each a_q by an integer verifier that is independent of the program that produced it;
- no uncontrolled floating-point approximation: Arb propagates rigorous rounding and truncation bounds.

A single-thread replay with a later engine version was bit-identical. The trust base is FLINT/Arb 3.3.1, GMP, MPFR, the compiler, the two program sources, and the hardware.

Theorem 3.4 (`R.4`; RANK)**.** $\operatorname{rank} E(\mathbf{Q}) = \operatorname{ord}_{s=1} L(E,s) = 2$.

Theorem 3.5 (`R.5`; computer-assisted)**.** *The least positive real period is*

$$\omega_1 = 0.028044901519436404448309402 8079 \pm 3.04 \cdot 10^{-32},$$

and $\Omega_E = 2\omega_1$. *The regulator of* P_1, P_2*, in the BSD normalization, satisfies*

$$17.79651747249034020748 < \mathrm{Reg} < 17.79651747249034039153,$$

and

$$0.99820315980984172437 < \Omega_E \,\mathrm{Reg} < 0.99820315980984173470.$$

The certificate evaluates the local heights with directed-rounding decimal arithmetic and an exact tail bound. It checks exactly that $2P$ lies in the identity component at every bad prime. It was replayed byte-identically at two precision settings.

Theorem 3.6 (`R.6`; computer-assisted)**.** $35.99427 < q < 36.00573$.

A much narrower floating-point estimate, $q = 36 \pm 3 \cdot 10^{-15}$, exists. It is evidence only, and is never used as an input to any proof here.

Theorem 3.7 (`R.7`; elementary)**.** *Full BSD for E holds if and only if* Ш *is finite of order* 9 *and* $q = 36$.

Proof. With $\prod c_\ell = 16$ and $\#E(\mathbf{Q})_{\mathrm{tors}} = 2$, the BSD formula reads $q = 16\,\#Ш/2^2 = 4\,\#Ш$. If it holds, $\#Ш = q/4$ is an integer in $(8.99856, 9.00144)$, hence equal to 9, and then $q = 36$. The converse is substitution. □

Theorem 3.7 describes the endpoint exactly; it supplies no method for either open assertion.

4. The primary components of Ш at $p \le 47$

4.1. **Method.** Let p be an odd prime at which E is ordinary: either good ordinary, or multiplicative. Let α be the unit root of $X^2 - a_p X + p$ when $p \nmid N$, and put $\alpha = a_p = \pm 1$ when $p \,\|\, N$. Let $\gamma = 1 + p$ and let ω be the Teichmüller character. Write $L_p = L_{p,\omega_1}$ for the cyclotomic series constructed from the ω_1-normalized plus symbols of Theorem 2.4. The full-real-period convention of Stein–Wuthrich [38] is $L_{p,\Omega_E} = L_{p,\omega_1}/2$. All displayed coefficient formulas use L_{p,ω_1}; both are integral and their unit status agrees at odd p. The construction is that of Mazur–Tate–Teitelbaum [30].

The depth-n approximation is the polynomial

$$P_n(T) = \sum_{a \in (\mathbf{Z}/p)^\times} \sum_{j=0}^{p^{n-1}-1} \Big(\alpha^{-n} \Big[\tfrac{\omega(a)\gamma^j}{p^n} \Big]^+ - \alpha^{-(n+1)} \Big[\tfrac{\omega(a)\gamma^j}{p^{n-1}} \Big]^+ \Big) (1+T)^j .$$

When $p \,\|\, N$ the second term is absent, because f is then already a U_p-eigenform with eigenvalue $\alpha = a_p$. Its low coefficients agree with those of $L_p(E,T)$ modulo an explicit power of p [38, Prop. 3.5]. The needed symbols are all p-integral. For each m, the cusps b/p^m lie in a single $\Gamma_0(N)$-class: that of 0 when $p \nmid N$, and that of $1/p$ when $p \,\|\, N$. The modular symbols therefore differ from one another by periods, and the Manin constant is a power of 2 (Theorem 2.4); see [38, §3] and the records cited in `S.3` and `S.7`.

Every coefficient below is therefore a *finite, predeclared, integer-weighted sum of modular symbols* $[b/p^2]^+$ (or $[b/81]^+$ at $p = 3$). Each symbol is computed as an Arb ball of radius below $1/2048$ and identified in $\frac{1}{1024}\mathbf{Z}$. The weights come from Fermat quotients: for $b = \omega(a)\gamma^j$ one has $j \equiv -(b^{p-1}-1)/p \pmod p$. The lower-level term drops out modulo p, because $\sum_{j<p} \binom{j}{2} = \binom{p}{3} \equiv 0$ for $p \ge 5$.

4.2. **Good ordinary primes.**

Lemma 4.1 (written proof applying published theorems; the argument of `S.7`)**.** *Let $p \ge 5$ be a prime of good ordinary reduction for E. If $C_p := [T^2]L_p(E,T)$ is a p-adic unit, then $Ш[p^\infty] = 0$.*

Proof. The residual image is full (Theorem 2.3), so Kato's divisibility [22], in the form of [38, Thm. 7.3], gives $L_p(E,T) = f_E(T)\, d(T)$ with $f_E, d \in \mathbf{Z}_p[[T]]$, where f_E is the characteristic power series of the dual Selmer group over the cyclotomic $\mathbf{Z}_p$-extension. Since $\operatorname{rank} E(\mathbf{Q}) = 2$, [38, Thm. 6.1] gives $\operatorname{ord}_T f_E \ge 2$.

A unit C_p forces $\operatorname{ord}_T L_p = \operatorname{ord}_T f_E = 2$ and makes the T^2-coefficient of f_E a unit. The equality clause of [38, Thm. 6.1] at good ordinary p (due to Schneider and Perrin-Riou) then shows that $Ш[p^\infty]$ is finite and that the p-adic height pairing is nondegenerate. Neither fact is assumed beforehand.

The leading-term formula now gives $v_p(\text{unit}) = v_p(\epsilon_p) + v_p(\#Ш[p^\infty]) + v_p(\mathrm{Reg}_\gamma)$, where the torsion and Tamagawa factors have been dropped because they are p-units. Here $\epsilon_p = (1-\alpha^{-1})^2$ is a unit: $\alpha \equiv a_p \pmod p$, a_p is even, and $|a_p| \le 2\sqrt{p}$ excludes $a_p \equiv 1 \pmod p$ for $p \ge 7$; at $p = 5$ the direct value $a_5 = -2$ gives the same conclusion.

Finally, $\mathrm{Reg}_\gamma \in \mathbf{Z}_p$. Multiplication by the p-unit $2\,\#E(\mathbf{F}_p)$ carries every rational point into the identity components at all finite places and into the formal group at p (Proposition 2.1(iv)). The canonical sigma-height formula [38, §4.1] then puts each height value in $p\mathbf{Z}_p$, while $v_p(\log_p \gamma) = 1$. Both valuations are nonnegative and they sum to zero, so $\#Ш[p^\infty] = 1$. □

The criterion is sufficient, not necessary. At $p = 17$ the normalized regulator has valuation one, so $C_{17} \equiv 0 \pmod{17}$ necessarily, and this criterion would need a depth-three coefficient (Remark 4.14); the prime 17 is settled instead by the singleton criterion of §4.5.

Theorem 4.2 (`S.5`, `S.7`, `S.11`, `S.13`; computer-assisted, published theorems applied)**.** $Ш[p^\infty] = 0$ *for* $p = 5, 7, 11, 13$*. The certified coefficients, in the* ω_1 *normalization, are:*

p	a_p	*certified weighted sum*	$C_p \bmod p$ *(*ω_1*)*	*full-period residue*
5	-2	*six-symbol value* 152	2	1
7	-2	-1931	1	4
11	-2	-190	8	4
13	-4	*group sums, total* -3950	2	1

Proof. Each residue is a unit, so Lemma 4.1 applies. At $p = 5$ the recorded argument uses the upper-bound direction of [38, Thm. 6.1] with the same inputs. □

Remark 4.3 (Certificates)*.* The certificates were built as follows.

- *Coefficients.* At $p = 7, 11, 13$ they were obtained from $6.9 \cdot 10^8$, $1.68 \cdot 10^9$ and (with reuse) about $2.3 \cdot 10^9$ independently verified prime coefficients, with Fourier truncations $1.56 \cdot 10^{10}$, $3.91 \cdot 10^{10}$ and $5.50 \cdot 10^{10}$ at denominator p^2.
- *Exact checks.* All plus and minus symbols at p and p^2 were identified. The two conjugation symmetries and every two-level Hecke relation hold exactly.
- *Independent evaluations.* Two exact downstream evaluations agree: one via Fermat quotients, one via the full depth-two measure modulo p^{20}.
- *The value at* 13. An external run first returned C_{13} as 7 or 8. Both values were traced to lattice-numerator scaling and to an omitted factor α^{-2}; after correction both give 2. The corrected, independently replayed downstream value is the one used here. The status-file custody discrepancy and the intact calculation bindings are distinguished in Appendix F.3.
- *Replay in the release.* The final step, from the certified symbol tables to the coefficients, is reproduced by an independent implementation of the formula of §4.1 (`tools/replay/coefficients_from_symbols.py`). It returns the residues $2, 1, 8, 2$ at $p = 5, 7, 11, 13$ and, at $p = 3$, the exact integers $a_{4,0} = 0$, $a_{4,1} = -54$ and $a_{4,2} = 15480$.
- *What was not done.* None of these certificates has had a second end-to-end implementation or independent human review.

Remark 4.4 (Formal check of the deduction)*. Formalized.* The deduction behind Lemma 4.1 and Theorem 4.2 is checked in Lean 4 (v4.28.0, Mathlib at tag v4.28.0), using only the axioms `propext`, `Classical.choice` and `Quot.sound`. The formal statements cover the following.

- *The criterion.* If $L_p = f_E\, d$ in $\mathbf{Z}_p[[T]]$, T^2 divides f_E, and $[T^2]L_p$ has nonzero reduction modulo p, then, under the hypotheses listed below, $Ш[p^\infty]$ is finite and trivial.
- *The certificate arithmetic at* $p = 5, 7, 11, 13$. This covers the point counts, $\alpha^{-2} \bmod p$ and the depth-two coefficients $[T^0]$, $[T^1]$, $[T^2]$ computed from the certified symbol tables for *arbitrary* lower-level symbols. It also checks the recorded weight tables entry by entry against their definition, and the exact integers 152, -1931, -190 and -3950.
- *Two general identities.* The lower-level term does not affect $[T^k]P_2 \bmod p$ for $k \le 2$ and $p \ge 5$, and the folded form of $[T^2]P_2$ holds for every prime $p \ge 5$.
- *A regression.* Omitting the factor α^{-2} (the error described in Remark 4.3 at $p = 13$) changes the residue at each of the four primes.

Every formal target was proved twice, in two separate runs. The author replayed each proof set from a fresh copy and checked that:

- the frozen statements were unchanged, and every target carried its frozen type;
- no unsafe construct was present;
- an independent compile against the pinned library reproduced the axiom report above.

Hypotheses, not formalized. The following enter as hypotheses: Kato's divisibility [38, Thm. 7.3]; the rank inequality, the good-ordinary equality clause and the leading-term formula of [38, Thm. 6.1]; the congruence of [38, Prop. 3.5]; the integrality of Reg_γ; and the unit property of ϵ_p and of the torsion and Tamagawa factors. (An abstract lemma shows that $1-\alpha$ is a unit when $a_p \not\equiv 1 \pmod p$; its application to the ϵ_p of E is not formalized.) The certified symbol tables enter as data, bound by SHA-256 to their certificate files.

Not checked. The imported theorems themselves, and the verification of their hypotheses for E, are not checked; nor are the numerical certificates of the symbol values. The formal check adds no arithmetic input and says nothing about $p \geq 17$ or the bad primes. The archive and the replay record are in `evidence/src/BSD_ARISTOTLE_FIN_VERIFICATION_2026-09-24/`.

4.3. **The prime 3.**

Theorem 4.5 (`S.3`; primary support: route K, Theorem 4.8; second route: computer-assisted, published theorems applied)**.** $Ш[3^\infty] \simeq (\mathbf{Z}/3)^2$.

Proof sketch of the second route. The primary proof is Theorem 4.8 together with Theorem 4.7. The second route, sketched here, uses Skinner's theorem. The reduction at 3 is nonsplit multiplicative ($\alpha = -1$, type I_2, $c_3 = 2$). The rigorous depth-four computation identifies $54+54$ plus/minus entries at denominator 81 and $18+18$ at denominator 27, hence 72 entries per sign across the two levels. It gives

$$a_{4,2} = 15480 = 2^3 \cdot 3^2 \cdot 5 \cdot 43 \equiv 9 \pmod{27}$$

in the ω_1 normalization; in the Néron normalization the value is $7740 \equiv 18$. By [38, Prop. 3.5], the quadratic coefficient a_2 of $L_3(E,T)$ satisfies $a_2 \equiv a_{4,2} \pmod{27}$, so $\mathrm{ord}_3 a_2 = 2$.

Skinner's theorem [36, Thm. A] applies to E at 3: the residual image is full, and $\bar\rho$ is ramified at 2243. It gives both divisibilities of the cyclotomic main conjecture. The Selmer local conditions and periods match integrally [17].

The order of inference matters. Finiteness comes first: the characteristic ideal has order exactly $2 = \mathrm{rank}\, E(\mathbf{Q})$, so the control argument gives corank 0 for $Ш[3^\infty]$. Only then is Jones's leading-term theorem [20, Thm. 3.1] used, in its forward direction. Its nondegeneracy hypothesis is supplied by an independent certificate that the 3-adic height determinant is nonzero with $\mathrm{ord}_3 \mathrm{Reg}_\gamma = 0$. The leading-term valuation then gives $\#Ш[3^\infty] = 9$, and the Cassels–Tate pairing excludes $\mathbf{Z}/9$. □

Theorem 4.6 (`S.3F`; published theorem applied to the certificate)**.** $Ш[3^\infty]$ *is finite. The proof uses no main conjecture.*

Proof. The reduction at 3 is nonsplit multiplicative, so E is not a Tate curve at 3, and $\alpha = a_3 = -1$. Kato's Theorem 18.4 [22] therefore applies at $p = 3$; it has no hypothesis $p \nmid N$ and no hypothesis on the image. It gives $\mathrm{corank}\, \mathrm{Sel}_{3^\infty}(E/\mathbf{Q}) \leq \mathrm{ord}_{s=1} L_{3,\alpha}(E)$, for the usual Selmer group. By the certificate in the proof of Theorem 4.5, $\mathrm{ord}_3 a_2 = 2$, so $a_2 \neq 0$ and the right-hand side is at most 2; orders of vanishing do not depend on the choice of period. (The identification of Kato's order of vanishing with that of $L_3(E,T)$ was checked by reading.) Since $\mathrm{rank}\, E(\mathbf{Q}) = 2$, the corank of $Ш[3^\infty]$ is 0. □

Theorem 4.7 (`S.3L`; exact computation)**.** $Ш[3] \supseteq (\mathbf{Z}/3)^2$*, that is,* $\#Ш[3] \geq 9$*. The proof uses no GRH and no main conjecture.*

Certificate. On the minimal model, with $P_1 = (-3031, 327384)$ and $P_2 = (-2394, 281520)$, consider the ternary cubics

$$C_1 : \ 61x^3 + 31x^2y + 15x^2z - 7xy^2 - 56xyz - 139xz^2 + 20y^3 + 75y^2z - 23yz^2 + 27z^3 = 0,$$

$$C_2 : \ 92x^3 + 49x^2y + 73x^2z - 61xy^2 + 50xyz + 31xz^2 - 35y^3 + 58y^2z - 58yz^2 - 31z^3 = 0.$$

Their invariants are $c_4 = 3169004368$ and $c_6 = 56243184546880$, those of E, so both have Jacobian E over $\mathbf{Q}$. Each is everywhere locally soluble. At $2, 3, 2243, 4547, 5843$ there are explicit points satisfying Hensel's condition. At every other prime the reduction is a smooth plane cubic, which has a point by the Hasse–Weil bound, and Hensel's lemma lifts it. Over $\mathbf{R}$ the degree is odd.

Let L be the degree-8 field generated by a 3-torsion point T of E. Through the Weil pairing with T, the classes $\kappa(P_1), \kappa(P_2), [C_1], [C_2] \in \mathrm{Sel}_3(E/\mathbf{Q})$ map to $L^\times/L^{\times 3}$, and their cubic residue symbols at 79 degree-one primes of L have $\mathbf{F}_3$-rank 4. So $\dim_{\mathbf{F}_3} \mathrm{Sel}_3(E/\mathbf{Q}) \geq 4$. Since $E(\mathbf{Q})[3] = 0$ and $\operatorname{rank} E(\mathbf{Q}) \leq 2$, $\dim_{\mathbf{F}_3} Ш[3] \geq 2$.

A standalone exact verifier recomputes each of these properties from the two cubics, the two points and a defining polynomial of L, in about 0.3 seconds, and rejects four negative controls. The search that found the cubics used class-group data conditional on GRH; the verifier does not. □

Theorem 4.8 (`S.3K`; written derivation; three Anthropic model readers (one cross-model: Fable 5.1; two independent fresh sessions: Opus 5.5) and one cross-family source reading (Codex, OpenAI) of D5/D6/D8 with the errata E1–E7, passing at the retained arithmetic-input and imported-theorem grades; not a full certificate replay; no human review)**.** $\#Ш[3^\infty] \leq 9$. *Hence, with Theorem 4.7,* $Ш[3^\infty] \simeq (\mathbf{Z}/3)^2$*, with no input from Skinner, Skinner–Urban or GRH.*

Sketch (route K). Let $T = T_3E$, let $\Lambda = \mathbf{Z}_3[[\Gamma]]$ for the cyclotomic $\mathbf{Z}_3$-extension of $\mathbf{Q}$, and let L_3 be the 3-adic L-function with $\alpha = a_3 = -1$ and the Néron differential. The 3-adic representation of E is surjective onto $\mathrm{GL}_2(\mathbf{Z}_3)$ (Elkies's criterion [16], computed), so Kato's image hypothesis (12.5.2) holds at 3.

The local construction. At the nonsplit multiplicative prime 3, T is an extension of $\mathbf{Z}_3(\eta)$ by $T^0 = \mathbf{Z}_3(1) \otimes \eta$, where η is the unramified quadratic character. Twisting Kobayashi's Coleman map for the Tate curve [27] by η gives an integral Coleman map

$$\mathrm{Col}\colon H^1_{\mathrm{Iw}}(\mathbf{Q}_3, T)/H^1_{\mathrm{Iw}}(\mathbf{Q}_3, T^0) \longrightarrow \Lambda.$$

It is an isomorphism. Indeed $H^2_{\mathrm{Iw}}(\mathbf{Q}_3, T^0) = 0$, because η stays nontrivial on the totally ramified cyclotomic tower; the formal group is free of rank one over each $\mathbf{Z}_3[\Gamma_n]$; and an explicit norm-compatible system has $\log c_0 = 3/2$, which generates $3\mathbf{Z}_3$. By Kato's explicit reciprocity law [22, 12.5(1)], compared character by character with the construction of [38, §3.2], Col sends Kato's zeta element to L_3, the Stein–Wuthrich series for a fixed period and lattice generator. Kato's own $L_{p\text{-adic},\alpha}$ is this series shifted by the cyclotomic character κ (a Tate twist: it is evaluated at $\kappa\chi$ where L_3 is evaluated at χ), not related to it by an involution, and the comparison with the series of the certificate holds up to a specified 3-adic unit. In particular $L_3 \in \Lambda$.

The global step. Kato's Poitou–Tate sequence [22, 14.9.3 and 17.10], arranged as in his §17.13, gives an exact sequence

$$0 \to H^1(T)/Z \to \Lambda/(L_3) \to X \to H^2(T) \to 0,$$

where X is the Pontryagin dual of the Selmer group over the cyclotomic $\mathbf{Z}_3$-extension. It is the classical Selmer group, because the local condition at 3 is the image of T^0, and it is primitive at 5843. Since $H^2_{\mathrm{loc}}(T) = 0$ at the nonsplit prime 3, Kato's Theorem 13.4(3), applied through 13.14, gives the integral inequality $\operatorname{length} H^2(T)_{\mathfrak{p}} \leq \operatorname{length} (H^1(T)/Z)_{\mathfrak{p}}$ at every height-one prime $\mathfrak{p}$ of Λ, including (3). Hence $f_E := \operatorname{char}_\Lambda X$ divides L_3 in Λ.

Assembly. Since $\operatorname{rank} E(\mathbf{Q}) = 2$, the control argument gives $\operatorname{ord}_T f_E \geq 2$. As $\operatorname{ord}_3 a_2 = 2$ (the certificate in the proof of Theorem 4.5), $L_3 = T^2 h$ and $f_E = T^2 g$ with $g \mid h$, so $\operatorname{ord}_3 g(0) \leq 2$ and $Ш[3^\infty]$ is finite. Jones's leading-term theorem [20, Thm. 3.1], in its forward direction, gives

$$g(0) = \mathrm{unit} \cdot \frac{\#Ш[3^\infty] \cdot \prod_\ell c_\ell \cdot \mathrm{Reg}_\gamma}{\#E(\mathbf{Q})[3^\infty]^2}.$$

Here $\mathrm{Reg}_\gamma = \det\langle\,,\rangle_\gamma$ is the γ-normalized regulator, which carries Jones's factor $(\log_3 \kappa(\gamma))^{-r}$ with $\mathrm{ord}_3 \log_3 4 = 1$. The height certificate gives $\mathrm{ord}_3 \mathrm{Reg}_\gamma = 0$. In general $\mathrm{ord}_3 \#Ш[3^\infty] \le 2 - \mathrm{ord}_3 \mathrm{Reg}_\gamma = 4 - \mathrm{ord}_3 \mathrm{Reg}_3$, where Reg_3 is the γ-free determinant. The γ-free determinant has valuation 2; division by $(\log_3 \kappa(\gamma))^2$ gives normalized valuation 0. If only valuation 0 for the γ-free determinant had been established, the resulting bound would be 81 rather than 9. The Tamagawa product 16 and $E(\mathbf{Q})_{\mathrm{tors}} = \mathbf{Z}/2$ are prime to 3, so $\#Ш[3^\infty] \le 9$. □

Remark 4.9 (Provenance at 3). Three qualifications apply to Theorem 4.5.

- *The second route is Skinner's theorem, imported as printed; two steps of its printed proof are not established for this curve.* For E at $p = 3$ the printed proof of [36, Thm. A] passes through Skinner–Urban's Theorem 3.6.1 [37], which uses [37, Prop. 12.3.6] twice. The printed proof of that proposition does not cover this case: at the forced inert prime 2243 the removed Euler factor $1 - 2243^{-2}$ has 3-adic valuation 1, Vatsal's theorem controls only the primitive factor, and (12.3.5.b) prints ℓ^{-1} where Skinner–Urban's own (3.4.5.b) gives q_v^{-1}. One use is repaired for this curve by a nonvanishing argument. The other needs [37, Prop. 13.4.1(ii)] for the data of E (MS-13.4.1(ii)); this is unproved, and it lies in the Selmer lower-bound direction. In the upper-bound direction, [37, Thm. 3.5.6] restates Kato's integral divisibility in the canonical-period normalization, whereas Kato proves it for a good differential; the comparison needed at the congruent forms of weight $k_m \ge 8 > 3$ (MS-B1) is in none of the sources we read. Skinner's printed theorem is not refuted.
- *The primary support does not depend on it.* Finiteness of $Ш[3^\infty]$ (Theorem 4.6), the lower bound $Ш[3] \supseteq (\mathbf{Z}/3)^2$ (Theorem 4.7) and the upper bound $\#Ш[3^\infty] \le 9$ (Theorem 4.8) are proved without [36]. Route K is therefore the primary support of Theorem 4.5, and [36, Thm. A], as printed, is kept as a second route with its two open steps for this curve, MS-13.4.1(ii) and MS-B1. Route K combines the integral Coleman map at the nonsplit multiplicative prime 3 (the η-twist of Kobayashi's Tate-curve map; this is the statement MS-R24) with Kato's Theorem 13.4(3) via 13.14 (the integral inequality at every height-one prime; $H^2_{\mathrm{loc}} = 0$ at nonsplit 3). Two other candidate routes are not needed: Ochiai's divisibility [31, Thm. 3 and Cor. 7.5] under the regularity statement MS-R3, and GRH together with one Cassels–Tate pairing value $\langle C_1, C_2\rangle$, which has not been computed.
- *The original written proof invoked a clause we do not use.* It cited the equality clause of [38, Thm. 6.1] at a multiplicative prime. The governing argument above instead uses finiteness first together with an independent height certificate, and imports no converse to Jones's theorem.

Remark 4.10 (Formal check at 3). *Formalized.* The deduction of Theorem 4.5 by the second route, from its inputs, is checked in Lean 4 (v4.28.0, Mathlib at tag v4.28.0), using only the axioms `propext`, `Classical.choice` and `Quot.sound`. It runs from the certified depth-four coefficient to $Ш[3^\infty] \simeq (\mathbf{Z}/3)^2$, in the order of inference above. The formal statements cover:

- the finite arithmetic at 3: the point count, the nonsplit reduction, $a_{4,\cdot} = (0, -54, 15480)$ and $\mathrm{ord}_3 a_{4,2} = 2$, the recorded weights, the U_3 relation with the independent denominator-27 table, depth three, and the normalization control;
- the criterion, in which finiteness comes from the control step, nondegeneracy from the independent height certificate, and the leading term from Jones's theorem in its forward direction;
- the Cassels–Tate step: a group of order 9 carrying a nondegenerate alternating pairing is $(\mathbf{Z}/3)^2$, and in general such a group has square order.

The proofs were produced in one run and replayed by the author from a fresh copy. The replay confirmed that the frozen statements and target types were unchanged, found no unsafe construct, and reproduced the axiom report above in an independent compile.

Hypotheses, not formalized. In the order of inference, the hypotheses are: Skinner's Theorem A [36] with the integral identifications of [17]; the control argument; the independent height certificate; Jones's Theorem 3.1 [20]; the congruence of [38, Prop. 3.5]; and the Cassels–Tate pairing. The certified symbol tables enter as data bound by SHA-256.

Not checked. The formal check does not cover the imported theorems, including the provenance limit of Remark 4.9, or the numerical certificates. Route K (Theorem 4.8) is not formalized. The archive and replay record are in `evidence/src/BSD_PERIMETER_Q36_2026-09-25/a5_replay/`.

4.4. **Consequences.**

Corollary 4.11 (`S.P`)**.** $Ш[30030^\infty] \simeq (\mathbf{Z}/3)^2$. *In particular* $\#Ш[30^\infty] = 9$.

Proposition 4.12 (`S.FIN`; written proof)**.** *If* Ш *is finite, then* $\#Ш = 9t^2$ *where every prime factor of* t *is at least* 53. *The next possible order after* 9 *is* $25281 = 9 \cdot 53^2$.

The bound is sharp under these inputs: $(\mathbf{Z}/3)^2 \oplus (\mathbf{Z}/53)^2$ carries an alternating perfect pairing [35]. The threshold has moved from 441 to 1089, 1521, 2601 and 25281 as components were settled; it should never be hard-coded. It uses Theorem 4.15, with that theorem's grades.

Proposition 4.13 (`S.U`; written proof; an equivalence)**.** Ш *is finite of order* 9 *if and only if* $\#Ш[n] < 2601$ *for every* $n \geq 1$.

Proof. If $\#Ш[n]$ is bounded by 2600, the increasing chain $Ш[m!]$ stabilizes, so Ш is finite; Proposition 4.12 then gives order 9. The converse is immediate. □

A uniform bound on $\#Ш[p]$ over primes p alone would not suffice. The group $(\mathbf{Z}/3)^2 \oplus (\mathbf{Q}_{17}/\mathbf{Z}_{17})^2$ shows why: full depth has to be controlled.

Remark 4.14 (The prime 17). At 17 one has $\operatorname{ord}_{17} \operatorname{Reg}_\gamma = 1$, so the depth-two test of Lemma 4.1 cannot succeed, and that cyclotomic route would need a depth-three coefficient $a_2 \not\equiv 0 \pmod{17^2}$, which is not computed. The requirement belongs to that criterion only: the prime 17 is settled by the singleton criterion below.

4.5. **The primes** $17 \leq p \leq 47$**: singleton certificates.** For a prime p and an auxiliary prime $\ell \equiv 1 \pmod p$, let g be the least primitive root modulo ℓ and put

$$C_k(p, \ell) = \sum_{a=1}^{\ell-1} \binom{\operatorname{ind}_g(a) \bmod p}{k} \left[\frac{a}{\ell}\right]^+,$$

with the ω_1-normalized plus symbols of Theorem 2.4. Replacing g by g^u multiplies C_2 by u^{-2} modulo p.

Theorem 4.15 (`S.17-47`; computer-assisted, a theorem of Ota applied as stated in its arXiv version)**.** $Ш[p^\infty] = 0$ *for* $p = 17, 19, 23, 29, 31, 37, 41, 43, 47$.

Proof. For each p one auxiliary prime ℓ was used, with $v_p(\ell - 1) = 1$, $\ell \nmid pN$ and $p \nmid \#E(\mathbf{F}_\ell)$. Every symbol $[a/\ell]^+$ was computed as an Arb ball and identified in $\frac{1}{1024}\mathbf{Z}$ by an exact rational check. In each case $C_0 = 0$ and $C_1 \equiv 0 \pmod p$, while $C_2 \not\equiv 0 \pmod p$:

p	17	19	23	29	31	37	41	43	47
ℓ	103	191	47	233	311	149	83	173	283
$C_2 \bmod p$	7	6	9	6	3	1	15	15	42

Ota's Theorem 6.4 [32] then gives $Ш[p] = 0$, with no finiteness assumed. A nonzero p-primary group would contain an element of order p, so $Ш[p^\infty] = 0$. Ota's criterion has no ordinarity hypothesis, so it applies at the supersingular primes 23 and 29. □

Remark 4.16 (Reviews and scope of Theorem 4.15). • *Reviews.* At $17, 19, 23, 29$ an independent audit by a language model of the same family as the tools that produced the runs replayed the frozen acceptance checks, recomputed the receipt roots, identified all 1140 intervals exactly and checked the prime-specific hypotheses; it is not a cross-family audit. At 37 and 41, and at $31, 43, 47$ (1528 intervals), the certificates were checked by models of a different family, within the replay limits recorded in the index. The large coefficient files were not rehashed in any of these reviews. No certificate has had human review.

- *Margins.* At 17 and 23 the ball radii are about 99% of the half-spacing $1/2048$. The identification is exact at the retained tail-bound grade, with little slack against an error in that bound.
- *Imports.* The rank, the residual image, the Manin constant with the $\frac{1}{1024}$ lattice, the interval engine and the coefficient corpus are retained inputs. Ota's theorems are applied as stated in arXiv:1509.00682v1, the version read at the source.
- *A residue observation, not a theorem.* With $R_{p,\ell}$ the tame regulator, the normalized residues $C_2(p,\ell)/((a_\ell - 2)R_{p,\ell})$ are consistent with a putative common integer congruent to 288 modulo $17 \cdot 19 \cdots 47 = 20{,}475{,}850{,}236{,}047$. No such common scalar has been constructed. Its conjecture-grade form is discussed after Research record 9.3.

Theorem 4.15 does not prove FIN.

5. The scalar endgame

Put $\varepsilon = 573/100000$. By Theorem 3.6, $|q - 36| < \varepsilon$.

Proposition 5.1 (`Q.1`; elementary)**.** *If $q = a/b$ in lowest terms with $1 \le b \le 174$, then $q = 36$. The bound is sharp for this interval: $36 + 1/175 \in (35.99427, 36.00573)$.*

Proof. If $q \neq 36$ then $|q - 36| \ge 1/b \ge 1/174 > \varepsilon$. □

Proposition 5.2 (`Q.2`; elementary)**.** *Let $f(X) = a\prod_j (X - q_j) \in \mathbf{Z}[X]$ be the primitive irreducible polynomial of $q = q_1$. If $\varepsilon\,|a| \prod_{j>1} |36 - q_j| < 1$, then $q = 36$.*

In particular, $q = 36$ as soon as q is a totally positive algebraic integer of degree at most two, with trace at most 246 in the quadratic case. The trace bound is sharp: the smaller root of $X^2 - 247X + 7595$ lies in the interval.

Proof. If $q \neq 36$ then $f(36)$ is a nonzero integer, so $1 \le |f(36)| = |a| \prod_j |36 - q_j| < \varepsilon |a| \prod_{j>1} |36 - q_j| < 1$, a contradiction. □

Neither arithmeticity hypothesis is known for q. Within the certified interval, each of them is *equivalent* to $q = 36$. A proof method would have to justify calling it easier.

Proposition 5.3 (`Q.3`; elementary; conditional)**.** *Suppose an independently constructed rational $B > 0$ satisfies both of the following:*

- *$q = 4B$;*
- *for every prime p, $Ш[p^\infty]$ is finite and $v_p(\#Ш[p^\infty]) \le v_p(B)$.*

Then $B = 9$, $\#Ш = 9$, and full BSD holds for E.

Proof. All valuations of B are nonnegative, so B is an integer, and the interval forces $B = 9$. The bounds kill every component except the 3-primary one, and $\#Ш[3^\infty] = 9$ by Theorem 4.5. □

Defining $B = q/4$ assumes the missing arithmeticity, and defining $B = \#Ш$ assumes FIN and the comparison. The mere existence of such a B is a reformulation of the endpoint.

Proposition 5.4 (`Q.4`; standard)**.** *Let $\alpha_f = 2\pi i f(z)\,dz$ and $\ell(z) = \log(-i\sqrt{N}z)$, and let γ be the ray from $i\infty$ to $i/\sqrt{N}$. Then*

$$\int_\gamma \alpha_f\, \ell^2 = \frac{L''(E,1)}{2}.$$

This identity, M1, holds before any finiteness statement. Several proposed arithmetic realizations of the left side are excluded in §10 (Obstructions 10.3–10.6). None of the exclusions disproves an arithmetic comparison obtained by some other construction.

6. Two quadratic twists

For a fundamental discriminant D with $\gcd(D, N) = 1$, write E^D for the quadratic twist. It has conductor D^2N, and its central values are computed from the same certified coefficients:

$$L(E^D, 1) = 2\sum_{n\geq 1} \frac{\chi_D(n)a_n}{n} e^{-2\pi n/(|D|\sqrt{N})} \quad (w(E^D) = +1),$$

$$L'(E^D, 1) = 2\sum_{n\geq 1} \frac{\chi_D(n)a_n}{n} E_1\left(\frac{2\pi n}{|D|\sqrt{N}}\right) \quad (w(E^D) = -1).$$

The tails are bounded by the same elementary bound used in Theorem 3.3.

Theorem 6.1 (`K.1`; computer-assisted, published theorems applied)**.** *The twist by $D = -23$ satisfies*

$$39.512236 \leq L'(E^{-23}, 1) \leq 40.182321.$$

Here the finite sum to $T = 3{,}456{,}106{,}495$ is $39.847278312318874 \pm 4.28 \cdot 10^{-16}$, and the tail is at most 0.335041797049. The root number of E^{-23} is -1, so its analytic rank is exactly one. By Gross–Zagier [18] *and Kolyvagin* [28]*, $\operatorname{rank} E^{-23}(\mathbf{Q}) = 1$ and $Ш(E^{-23}/\mathbf{Q})$ is finite. Hence $\operatorname{rank} E(\mathbf{Q}(\sqrt{-23})) = 3$.*

Theorem 6.2 (`K.2`; computer-assisted, published theorems applied)**.** *The twist by $D = -7$ satisfies*

$$9.054319 \leq L(E^{-7}, 1) \leq 9.054421.$$

Here $T = 2{,}013{,}265{,}919$, and the margin over the tail bound is $1.8 \cdot 10^5$. The root number is $+1$, so the analytic rank is zero, and $\operatorname{rank} E^{-7}(\mathbf{Q}) = 0$ with $Ш(E^{-7}/\mathbf{Q})$ finite. Hence $\operatorname{rank} E(\mathbf{Q}(\sqrt{-7})) = 2$, and the two ranks are $(r^+, r^-) = (2, 0)$.

Remark 6.3. The two fields play different roles.

- In $\mathbf{Q}(\sqrt{-23})$ all bad primes split: this is the indefinite, Heegner setting, of rank three.
- In $\mathbf{Q}(\sqrt{-7})$ the primes 3, 2243 and 5843 are inert and 2, 4547 split. The definite quaternion algebra ramified at $\{3, 2243, 5843, \infty\}$ then carries the Gross points used in §7.

Finiteness of the twists' Tate–Shafarevich groups gives no information about $Ш(E/\mathbf{Q})$ by itself.

The $D = -7$ certificate has further support:

- an independent small-T implementation (a sieve and exact multiplicativity) agrees to $1.8 \cdot 10^{-15}$;
- it passed an external re-audit of unit hashes, segment roots and exact rational recombination.

7. Sufficient criteria at good primes

Each result in this section is a *sufficient* criterion at a single prime. A failed criterion proves nothing, and no criterion here holds uniformly in p.

7.1. **Cyclotomic criteria.**

Criterion 7.1 (B.1; written proof applying published theorems)**.** *For every good ordinary prime $p \geq 5$: if $C_p = [T^2]L_p(E,T) \not\equiv 0 \pmod{p}$, then $Ш[p^\infty] = 0$.*

This is Lemma 4.1, stated for all good ordinary primes; for this curve every other hypothesis is automatic. The coefficient C_p mod p is a finite weighted sum of the symbols $[b/p^2]^+$, and it is computable with the engine of Appendix B. The Fourier cutoff grows roughly like $p^2\sqrt{N}$: it was $5.5 \cdot 10^{10}$ at $p = 13$.

Criterion 7.2 (B.2; written proof applying published theorems)**.** *Let $p \geq 5$ be a good supersingular prime, so $a_p = 0$. If the X^2-coefficient of the depth-one Mazur–Tate element θ_1 is nonzero modulo p, then $Ш[p^\infty] = 0$. That coefficient is a finite weighted sum of the symbols $[a/p^2]^+$.*

The proof combines three published inputs:

- Pollack's congruence $\theta_1 \equiv \widetilde{\omega}_1^+ L_p^+ \pmod{\omega_1}$, which has no cyclotomic factor at $n = 1$ [34, Prop. 6.18], [24, Prop. 1.11];
- Kobayashi's integral divisibility $(L_p^+) \subseteq \mathrm{char}(X^-)$ [26];
- level-zero control and the absence of finite Λ-submodules [25].

The criterion passed a hostile automated audit with corrections to constants and sign labels, none of which affects the implication. Some primary results were read through the restatements identified in the claim index. For this curve the estimated costs are 270 and 460 core-hours at $p = 23$ and 29; neither computation has been run.

7.2. **Anticyclotomic forward bridges.** Let $K = \mathbf{Q}(\sqrt{-7})$. By Theorem 6.2, $\mathrm{rank}\, E(K) = 2$ with $(r^+, r^-) = (2, 0)$. The primes $3, 2243, 5843$ are inert in K, which puts us in the definite setting of Bertolini–Longo–Venerucci [5] with $N^- = 3 \cdot 2243 \cdot 5843$ and $N^+ = 2^3 \cdot 4547^2$.

For a good prime $p \geq 5$ with $p \nmid 7N$, let Q_1^{Gr} be BLV's traced Gross point of conductor p^2, and let $\bar{\psi}$ be the reduction of the (primitive) Jacquet–Langlands transfer of f. Put

$$M_i(p) = \sum_{j=0}^{p-1} j^i \, \bar{\psi}(\gamma^j Q_1^{\mathrm{Gr}}) \in \mathbf{F}_p.$$

Theorem 7.3 (B.4; written proof, algebraic core Lean-checked, published theorems applied; conditional)**.** *Let $p \geq 5$ be a good supersingular prime with $p \nmid 7N$. Then:*

(i) $M_0(p) = M_1(p) = 0$.
(ii) $\bar{b}_2 = M_2(p)/2$, *where b_2 is the T^2-coefficient of BLV's primitive minus square-root series F^-.*
(iii) If $M_2(p) \neq 0$, *then* $Ш[p^\infty] = 0$.

Summary; complete argument in Appendix E. If $\overline{F^-} \neq 0$, put $d = \mathrm{ord}_T(\overline{F^-})$. This implies $\mu = 0$ and $\lambda = d$. The argument in this case has three steps.

1. *No finite submodule.* Equality in BLV's Theorem 7.1 at a character of infinite order shows that the dual signed Selmer module X has no finite Λ-submodule.
2. *Balanced ranks.* A τ-equivariant freeing-set argument (Poitou–Tate, BLV §5) gives $\mathrm{rank}_{\mathbf{Z}_p} X^+ = \mathrm{rank}_{\mathbf{Z}_p} X^- = d$.
3. *Conclusion.* Nonvanishing of $M_2(p)$ gives $d = 2$. With $\mathrm{rank}\, E(\mathbf{Q}) = 2$, the surjection $X^+ \twoheadrightarrow \mathrm{Sel}(E/\mathbf{Q})^\vee \twoheadrightarrow \mathbf{Z}_p^2$ is then an isomorphism, so $Ш[p^\infty]^\vee = 0$.

The same rank count forces $d \geq 2$, so the constant and linear coefficients vanish. If $\overline{F^-} = 0$, those coefficients vanish directly. The first-layer packet formula then gives $M_0 = M_1 = 0$ and $\bar{b}_2 = M_2/2$. □

Remark 7.4 (What is checked, and how). *Formalized.* The algebra of steps 1–3 is checked in Lean 4 (v4.32.2, Mathlib at tag v4.32.2): zero errors and zero warnings, using only the axioms `propext`, `Classical.choice` and `Quot.sound`. A rebuild made while this release was assembled produced output identical to the recorded build log.

Hypotheses, not formalized. Two groups of inputs enter the Lean statement as hypotheses:

- the four arithmetic inputs: control, exact length at one character, the graded free presentation with the base-layer eigenspace equality, and rank $E(\mathbf{Q}) = 2$;
- four structural facts about Λ-modules.

Two modeling steps (eigenspace exactness in characteristic $\neq 2$, and Pontryagin duality) are stated but not formalized.

What the formalization sharpened. The control input is used only as a τ-compatible surjection; only the minus part of the presentation equality is used; and the separate $\mu = 0$ hypothesis is unnecessary when one assumes finite $d = \mathrm{ord}_T(\overline{F^-})$. Finiteness of the usual Iwasawa λ alone does not imply $\mu = 0$ (for example, pT^2 has $\mu = 1$, $\lambda = 2$).

Source checking and imports. The BLV citations were checked at the source (arXiv v2). BLV's own imports (Fouquet–Wan and others) were not read.

Theorem 7.5 (`B.3`; written proof, published theorems applied; conditional)**.** *Let $p \geq 5$ be a prime of good ordinary reduction with $p \nmid 7N$, and let $c_2(p)$ be the I^2/I^3-coefficient of the stabilized Gross-point square-root theta element. If $c_2(p) \in \mathbf{Z}_p^\times$, then $Ш[p^\infty] = 0$.*

The records contain two proofs.

- *The frozen route.* The lower bound $\mathrm{ord}_I \mathcal{L}_p \geq 2\max(r^+, r^-)$ [4, Cor. 3] holds unconditionally, by Theorem 6.2. A unit $c_2(p)$ forces equality, and BLV's equality of characteristic ideals holds in the ordinary case. The derived-height theory of Bertolini–Darmon [3] then gives $\mathrm{Div}\, Ш(E/K)[p^\infty] = 0$ with the exponent bound $v_p \# Ш(E/K)[p^\infty] \leq 2c$, where p^c is the exponent of $B = Ш(E^{-7}/\mathbf{Q})[p^\infty]$. Because $Ш(E/K)[p^\infty] \simeq Ш(E/\mathbf{Q})[p^\infty] \oplus B$ for odd p, and the Cassels–Tate pairing on the finite group B forces $\#B \geq p^{2c}$, the summand $Ш(E/\mathbf{Q})[p^\infty]$ vanishes.
- *The A3 route.* The argument of Theorem 7.3, with trivial sign, applies verbatim at ordinary p. It uses neither derived heights nor the twist certificate, and its algebraic core is the same Lean development.

The two routes differ in how much has been checked. The frozen route rests on several steps read only by two external automated audits: the BLV–BD module identification, [3, Thms. 2.18(5), 2.23, Lemma 3.1], and the odd-p descent. The records designate it as canonical, and record the second route without substituting it (Appendix C).

Corollary 7.6 (`B.5`; conditional)**.** *For every good prime $p \geq 5$ with $p \nmid 7N$, ordinary or supersingular, $M_2(p) \neq 0$ implies $Ш[p^\infty] = 0$. At ordinary p, given $M_1 = 0$, one has $\bar{c}_2 = c_{p-3}/2 = -M_2/2$, where c_i are the Lagrange coefficients of $j \mapsto \bar{\psi}(\gamma^j Q_1^{\mathrm{Gr}})$ on $\mathbf{F}_p$.*

Remark 7.7 (No prime is gained). Nonvanishing of $M_2(p)$ is open at every prime. We know of no way to evaluate $M_2(p)$ for this curve. The form $\bar{\psi}$ lives on the definite Shimura set attached to the quaternion algebra ramified at $\{3, 2243, 5843, \infty\}$ with Eichler level $N^+ = 2^3 \cdot 4547^2$, whose mass is

$$\frac{1}{12} \prod_{q \mid N^-} (q-1) \cdot \psi(N^+) \approx 5.4 \cdot 10^{14}.$$

This mass is a scale estimate for full enumeration, not the exact number of classes or a lower bound for every evaluation algorithm. Full enumeration exceeds the project's resources; no compressed evaluator of the required packet has been supplied. The conditional theorems remain

single-prime criteria and could become computationally useful if such an evaluator is found. The currently implemented individual-prime tools are Criteria 7.1 and 7.2.

Remark 7.8 (Stabilization). For $p \geq 5$ the norm element $\sum_{i<p}(1+t)^i$ lies in $p\Lambda + I^3$ (`B.15`). So the α-stabilization correction in the ordinary theta measure is invisible modulo (p, I^3), which is exactly the quotient detecting the degree-two witness.

8. The bad primes 2243, 5843 and 4547

The good-ordinary version of Kato used here, in [38, Thm. 7.3], is not the input for these bad primes. At the multiplicative primes we instead use [36, Thm. A], as at $p = 3$; no blanket exclusion of all other versions of Kato is asserted.

8.1. **The split multiplicative prime** 2243**.** Let $p = 2243$. Here $\alpha = a_p = +1$ and the reduction type is I_1. The p-adic L-function has an exceptional zero, so $\mathrm{ord}_T L_p \geq r+1 = 3$. For $b \in (\mathbf{Z}/p^2)^\times$ put

$$\ell(b) \equiv -\frac{b^{p-1}-1}{p} \pmod{p},$$

and define the integer

$$K = 1024 \sum_{b \bmod \pm 1} \mathrm{bal}_p\Big(2\binom{\ell(b)}{3}\Big)\Big[\frac{b}{p^2}\Big]^+,$$

where bal_p is the balanced residue modulo p. We use the ω_1-normalized version of the Stein–Wuthrich construction; the full-period series is half this series, a unit factor at p.

Theorem 8.1 (`B.6`; conditional)**.** *If $K \not\equiv 0 \pmod{2243}$, then $Ш[2243^\infty] = 0$, given the mathematical inputs D1–D7 below. If $K \equiv 0 \pmod{2243}$, nothing follows.*

input	content	status
D1	Hypotheses of [36, Thm. A] at 2243: ordinary, residually irreducible, ramified at a prime $q \,\|\, N$ ($q = 3$ or 5843)	verified; theorem imported as published
D2	$\mathrm{char}(X_{\mathrm{Gr}}) = T \cdot \mathrm{char}(X_{\mathrm{cl}})$ in the split case	**discharged** at source level: [36, §3.2] together with a written matching of Selmer local conditions (strict condition = Kummer image via the Tate curve at p; trivial at $\ell \neq p$ over $\mathbf{Q}_\infty$)
D3	Comparison of canonical and Néron periods at $p \mid N$	as published; the internal provenance limit of Remark 4.9 recurs
D4	Finiteness first: $\mathrm{corank}\,\mathrm{Sel}_{p^\infty}(E/\mathbf{Q}) \leq \mathrm{ord}_T f_E$	elementary control argument
D5	Nondegeneracy of the split height	**certified** by an independent computation (Theorem 8.2)
D6	Jones's split leading-term formula [20, Thm. 3.1], forward direction; it needs $\log_p q_E \neq 0$	read at the source; $\ell_\gamma = \log_p(q_E)/(\mathrm{ord}_p(q_E)\log_p(1+p)) \equiv 1799 \pmod{2243}$, a unit, computed twice independently
D7	[38, Prop. 3.5] and the symbol lattice (Theorem 2.4)	read; lattice derivation in Appendix D

Certificate production (formerly D8). Evaluator reliability is needed to certify a computed K, not to define K or prove the implication above. The retained pipeline still needs its restart-isolation defect F4 corrected and the repaired coefficient verifier installed before production.

Theorem 8.2 (`B.6`, input D5; computer-assisted, published height formula imported)**.** *Let H be the matrix of the corrected split-multiplicative height pairing* [38, §4.2] *(Werner's correction* [41]*) on the basis P_1, P_2 of the minimal model. Then*

$$H = 2243\begin{pmatrix}2127 & 1294\\ 1294 & 764\end{pmatrix} + O(2243^2), \qquad \det H = 2181 \cdot 2243^2 + O(2243^3).$$

In particular H is nondegenerate, and with $\kappa(\gamma) = 1 + 2243$ the normalized regulator satisfies $\mathrm{Reg}_\gamma \equiv 2181 \pmod{2243}$, a unit.

The certificate works as follows.

- *Arithmetic.* It uses Python's standard-library integers and fractions only; no PARI or Sage height function is called.
- *Truncation.* The Bernardi series and the canonical correction are accounted for to the required precision: every omitted term of $\log(\sigma(t)/t)$ lies in $p^2\mathbf{Z}_p$.
- *The Tate parameter.* q_E is computed modulo p^3 from Δ/c_4^3.
- *Global denominators.* The integer $e(Q)$ with $x(Q) = a/e(Q)^2$ in lowest terms, including its part prime to p, is obtained through division polynomials, with an explicit check that no bad prime cancels.
- *Verification.* A separate division-free verifier and a Jacobian-coordinate multiplication agree. The parallelogram identity, scaling, and a unimodular change of basis are checked.

The PARI residue was consulted only after the computation, and it agrees. The published height formula is imported for this computation; its use in D6 additionally retains the published leading-term comparison and the stated normalization checks.

Remark 8.3 (Why D5 mattered). This proof uses only the forward leading-term implication of [20, Thm. 3.1], with finiteness and nondegeneracy supplied independently. It does not require a decision about every converse attributed in the later literature. Finiteness comes from D4, nondegeneracy from Theorem 8.2, and then Jones's forward theorem applies. This is exactly the shape of the proof at $p = 3$.

Proposition 8.4 (`X.15`; written proof and exact computation)**.** *Write $b = \omega(a)(1+p)^j$, so that $j = \ell(b)$. Let $Y_j = \sum_a 1024[b/p^2]^+$ be the column sums of the depth-two symbol matrix, and put $\theta(T) = \sum_j Y_j(1+T)^j$. Then:*

(i) $K \equiv c_3 := [T^3]\theta \pmod p$.
(ii) The U_p trace relations fix only the row sums, hence only $c_0 = 0$.
(iii) The Atkin–Lehner involution W_Q ($Q = N/p$) gives $[u_b/p^2]^+ = -[b/p^2]^+$ with $u_b = (Qb)^{-1}$. It maps column j to column $-1989 - j$, where $1989 \equiv \ell(Q) \pmod p$.
(iv) $(1+T)^{1989/2}\theta$ is odd in the uniformizer $W = T(1+T)^{-1/2}$. Consequently $c_0 = 0$, $c_2 = -995c_1$ and $c_4 = 1247c_3 + 511c_1$, while c_3 is not determined by c_0, c_1, c_2: the rank of these coefficient functionals over $\mathbf{F}_{2243}$ is 2, not 1.

Consequently no Manin, Hecke or Atkin–Lehner relation among those examined expresses K through cheaper data. We examined:

- the U_p trace relations and the Atkin–Lehner involutions W_Q, W_p, W_N;
- $\Gamma_0(N)$-loops;
- the operators T_ℓ and U_ℓ;
- Hida-family relations and Galois conjugation;

- aggregation of the certificate sum;
- Kolyvagin-type auxiliary denominators.

Each of the alternatives raises the analytic scale (by factors of 16 up to about $5 \cdot 10^{11}$) or needs a modular-symbol space of dimension 10^{15}–10^{16}. This is an obstruction for the relations examined, not a lower bound for every algorithm.

The cost of computing K directly is set by the analytic conductor of the order-p twists. The Fourier cutoff is $3.912 \cdot 10^{13}$, about $1.1 \cdot 10^5$ core-hours. The operational cost models remain in the repository; they are not mathematical hypotheses. A future computation carries a free consistency check: $c_1 \equiv 0 \pmod{2243}$ is forced by the order of vanishing but not by the pairing, so a failure detects an error. No probability of detecting arbitrary incorrect output is asserted.

8.2. **The nonsplit multiplicative prime** 5843**.**

Criterion 8.5 (`B.7`; conditional)**.** *At* 5843 *(type* I_2*,* $a_p = -1$*), one depth-two coefficient modulo* p *is a sufficient certificate, through Skinner's theorem and Jones's nonsplit formula as at* $p = 3$*.*

The Fourier cutoff is $1.687 \cdot 10^{14}$, and the estimated cost is about $6.3 \cdot 10^5$ core-hours. It has not been computed.

8.3. **The additive prime** 4547**.**

Remark 8.6 (`B.8`; retained project boundary)*.* At 4547 (type III, potentially supersingular, U_p-eigenvalue 0), the retained records supply no cyclotomic chain of a p-adic L-function, a main conjecture and a control theorem in the required form. The retained records contain a two-prime Kurihara witness, whose existence at an additive prime is conjectural; its smallest Kolyvagin pair is about $8.1 \cdot 10^{15}$, and its direct implementation cost is prohibitive within the project budget. The quadratic singleton criterion (Criterion 8.7 below) is a further sufficient criterion at 4547: additive reduction does not block its unramified auxiliary extension. No witness has been evaluated at 4547. This is neither an impossibility theorem for that prime nor an exclusion of new algorithms.

8.4. **A quadratic singleton criterion at the bad primes.**

Criterion 8.7 (`B.18`; conditional; a preprint theorem applied)**.** *Let* $p \in \{2243, 4547, 5843\}$ *and let* ℓ *be a prime with* $\ell \equiv 1 \pmod{p}$*,* $v_p(\ell - 1) = 1$ *and* $\ell \nmid pN$*. Define* $C_k(p, \ell)$ *as in §4.5, over the full label set. If* $C_0 \equiv C_1 \equiv 0$ *and* $C_2 \not\equiv 0 \pmod{p}$*, then* $Ш[p^\infty] = 0$*.*

This is a written corollary of Theorem 1.4(b) of Bullach–Honnor [7], with the rank, residual-image and symbol-lattice inputs retained. Their theorem bounds a Fitting ideal of the dual p-primary Selmer group over the degree-p subfield of $\mathbf{Q}(\mu_\ell)$. A nonzero quadratic coefficient then allows at most two generators, and Nakayama's lemma kills the dual of $Ш[p^\infty]$; no finiteness is assumed. The normalizations differ by an exact factor: Bullach–Honnor divide by the total real period $2\omega_1$, so $\theta_{\mathrm{BH}} = \theta_{\mathrm{pipeline}}/2$, and no further Manin or exceptional-zero factor enters. Ota's hypotheses are not claimed at these primes; this is a separate theorem profile. No coefficient has been evaluated at any of the three primes, and a zero value would prove nothing. The cost models are in §10.

9. Equivalences and reductions

This section summarizes archived reductions. The research-record entries use definitions and hypotheses supplied in their linked claim-index evidence; they are not standalone theorem statements or proofs in this manuscript. They are not used to prove the main results. Some restate parts of the open problem in other terms. An equivalence earns no reduction in difficulty by itself. We record these results because they fix the exact shape of the remaining quantifiers.

Research record 9.1 (B.9; written proof from published and preprint sources; an equivalence)**.** Let $S_0 = \{2, 3, 5, 2243, 4547, 5843\}$. For every good ordinary prime $p \notin S_0$,

$$h_p = 0 \iff Ш[p^\infty] = 0,$$

where h_p is the witness depth of Ota's theta elements [32] over admissible pairs of auxiliary primes.

The forward direction is Ota's Theorem 6.4. The converse combines three inputs:

- Castella–Sano [10], whose Theorem B produces a p-unit Kurihara number;
- C.-H. Kim [23], Theorem 1.10, which localizes the Selmer group at the minimal pair;
- a normalization check between the period conventions.

Uniform primitive witnesses on the ordinary tail are exactly uniform vanishing there. The theorem gives no converse at positive depth or at supersingular primes.

Research record 9.2 (B.10; written proof; an equivalence)**.**
Let $p \geq 7$ with $p \notin \{19, 59, 487, 2243, 4547, 5843\}$, and put

$$\mathcal{C}_p = \mathrm{Hom}_{\mathrm{GL}_2(\mathbf{F}_p)}(\mathrm{Cl}(\mathbf{Q}(E[p]))/p,\ E[p]).$$

Then $Ш[p] = 0$ if and only if $\dim_{\mathbf{F}_p} \mathcal{C}_p \leq 2$. Moreover $\dim \mathcal{C}_p \geq 1$.

The excluded primes limit where this simplified reduction applies. They are *not* a bound on the primes dividing $\#Ш$.

Research record 9.3 (B.11; written proof, finite algebra Lean-checked; conditional)**.** *LINE* is the hypothesis that one rational scalar a/b relates the tame leading terms δ_n and the regulator terms ρ_n at all required auxiliary pairs and primes, by $b\delta_n = a\rho_n$. Under LINE, the primary components at the covered good primes are finite, and they vanish away from the support of the numerator.

Witness content weakens LINE to the existence of a nonzero integer $A \in J_E$, and the same conclusion holds under it.

The finite algebra of both implications is formalized in Lean (the frozen archives A1 and A2). The arithmetic antecedents are not proved: neither the cross-prime scalar of LINE nor the nonzero integer A is known to exist.

LINE is related to the residue observation of Remark 4.16, but it is not equivalent to it. LINE needs a common rational comparison on core pairs at every p-power level, with auxiliaries that carry p-power point torsion; the observation concerns nonvanishing modulo p at selected auxiliaries (Research record 9.7). Neither is known to imply the other. At the grade of a conjecture (Conjecture 2.3 of Burns–Kurihara–Sano [8], in the production conventions) the constant of the tame comparison is

$$K_E = \frac{4\, c_\infty \, \#Ш \, \mathrm{Tam}}{\#E(\mathbf{Q})^2_{\mathrm{tors}}}, \qquad c_\infty = \#\pi_0(E(\mathbf{R})).$$

It matches exactly on six rank-two curves with trivial Ш: 68 of 68 rows with nonzero tame regulator, and 7 of 7 rows where both sides vanish. The factor $\#Ш$ was tested on the curve 194040co6, whose analytic order of Ш is 4: 17 of 17 rows match, and its isogenous curves with trivial Ш match 40 of 40. Every constant there is a power of 2. No rank-two curve of conductor below 500,000 has odd analytic order of Ш other than 1, so an odd factor such as the 9 expected here cannot be tested on such curves. For E one has $K_E = 32\,\#Ш$, and $288 = 32 \cdot 9$ is consistency evidence, at conjecture grade, for $\#Ш \equiv 9$ modulo each observed prime. It says nothing about q.

Proposition 9.4 (B.12; standard deduction with an exact computation)**.** *(i) For $p \in \{2,3,5,7,11\}$, the rational p-adic Selmer group is*

$$\mathbf{Q}_p\delta_p(P_1) \oplus \mathbf{Q}_p\delta_p(P_2).$$

(ii) For $p = 5, 7, 11$, one has $\mathrm{Sel}_{p^n}(E/\mathbf{Q}) \simeq (\mathbf{Z}/p^n)^2$ for all n, and

$$\ker(\mathrm{loc}_p) = \mathbf{Q}_p\big(\log_p(P_2)\,\delta_p(P_1) - \log_p(P_1)\,\delta_p(P_2)\big).$$

(iii) At $p = 11$, with $S = \{\infty, 2, 3, 11, 2243, 4547, 5843\}$, the kernel of loc_S on $\mathrm{Sel}_{11}(E/\mathbf{Q})$ is the line $\mathbf{F}_{11}\,\delta_{11}(2P_1 - P_2)$.

Research record 9.5 (B.13; written source composition; conditional)**.** Let $p \geq 5$ be a good ordinary prime with $p \notin \{19, 59, 487\}$ and $p \nmid \#\widetilde{E}(\mathbf{F}_p)$. Combining Burungale–Castella–Kim [9], Kataoka–Sano and Macias Castillo–Sano [29] gives two things over a suitable imaginary quadratic field:

- a base-layer Heegner Stark-system basis;
- Fitting-ideal identities for the classical Selmer group at every depth.

For $K = \mathbf{Q}(\sqrt{-23})$ (Theorem 6.1) the identity reads

$$\mathrm{Fitt}^{\,i}\big(Ш(E/K)[p^n]^*\big) = I_{i+3}(\epsilon^{\mathrm{Hg}}_{n,0}),$$

at $p = 13, 31, 41, 47, 71, 73, 101, 127, 139, 151$ among the primes $p \leq 160$.

The ideals I_{i+3} are not evaluated at any prime, and the Heegner Λ-index does not determine them (an explicit counterexample is recorded). Moreover, $I_3 = \mathbf{F}_{13}$ would require $Ш(E^{-23}/\mathbf{Q})[13] = 0$ as well as the seed's 13-part. The twist part is not supplied, so no unit specialization is known at 13 (see the withdrawn claim W.1).

Lemma 9.6 (B.14; Lean-checked)**.** *Let A be a fixed integer matrix. Suppose that for every p, $I_3(p) = \mathbf{F}_p$ holds if and only if some $k \times k$ minor of A is nonzero modulo p, and that this holds for at least one p. Then $C_1 = |\gcd(k \times k \text{ minors of } A)|$ is nonzero, and $I_3(p) \neq \mathbf{F}_p$ implies $p \mid C_1$.*

No such integral model has been constructed, so Lemma 9.6 currently has no instance.

Research record 9.7 (B.16; written proof applying published constructions)**.** Let $S_0 = \{2, 3, 5, 2243, 4547, 5843\}$. For every prime $p \notin S_0$ there are infinitely many primes $\ell \nmid pN$ with

$$\ell \equiv 1 + p \pmod{p^2}, \qquad \bar{\rho}_{E,p}(\mathrm{Frob}_\ell) = -1, \qquad R_{p,\ell} \neq 0 \text{ in } \mathbf{F}_p,$$

where $R_{p,\ell}$ is the normalized determinant of the tame pairing on the basis of Theorem 3.1. In particular $v_p(\ell - 1) = 1$ and $\#E(\mathbf{F}_\ell) \equiv 4 \pmod{p}$.

The proof fixes one Poincaré biextension over $\mathbf{Q}$ before p or ℓ is chosen, following Tan [40] and Jossen [21], shows that the relevant Kummer image is full, and applies Chebotarev's theorem. It is existential and gives no bound for the smallest ℓ. It does not identify $C_2(p, \ell)$ with $R_{p,\ell}$, so it proves nothing about Ш.

Research record 9.8 (B.17; written proof from preprint sources)**.** Let p be a prime outside $\{2, 3, 5, 19, 59, 487, 2243, 4547, 5843\}$, let m be a squarefree product of primes $\ell \nmid pN$ with $\ell \equiv 1 \pmod{p}$ and $p \nmid \#E(\mathbf{F}_\ell)$, and let I be the augmentation ideal of $\mathbf{Z}_p[\mathrm{Gal}(F/\mathbf{Q})]$ for the corresponding abelian p-extension F. Put $s_p = 2 + \operatorname{corank} Ш[p^\infty]$ and $A_p = H^2_f(\mathbf{Q}, T_pE)_{\mathrm{tors}}$. If $s_p > 2$, the degree-two witness $t_{p,m}$ is 0. If $s_p = 2$, then $Ш[p^\infty]$ is finite, $\#A_p = \#Ш[p^\infty]$, and

$$t_{p,m} = u_{p,m}\, x_{\mathbf{Q},\Sigma}\, \#A_p\, \det(B_m) \quad \text{in } I^2/I^3,$$

where $u_{p,m}$ is a unit, $x_{\mathbf{Q},\Sigma}$ is the coordinate of the determinantal Kato element and $\det(B_m)$ is the Bockstein determinant.

The inputs are the determinant constructions of Burns–Kurihara–Sano [8], arithmetic Selmer duality and a Bockstein theorem, all imported. This is an identity in a finite quotient: a sufficiently divisible coefficient is 0, and its depth is not a finite truncation endpoint. Nothing here bounds the multiplier $x_{\mathbf{Q},\Sigma}\#A_p\det(B_m)$ uniformly in p. A nonzero singleton witness in this range forces both $Ш[p^\infty]=0$ and primitivity of the Kato element; the range excludes 19.

Research record 9.9 (`B.19`; written proof from published and preprint sources)**.** Let $p\notin B^*=\{2,3,19,59,487,2243,4547,5843\}$, and let $\ell\neq\ell'$ be primes with $\gcd(\ell\ell',pN)=1$, $\ell\equiv\ell'\equiv 1 \pmod p$, $p\nmid \#E(\mathbf{F}_\ell)\,\#E(\mathbf{F}_{\ell'})$ and $R_{p,\ell}R_{p,\ell'}\not\equiv 0 \pmod p$. With one period, one saturated basis and matching generators,

$$\frac{C_2(p,\ell)}{(a_\ell-2)R_{p,\ell}}=\frac{C_2(p,\ell')}{(a_{\ell'}-2)R_{p,\ell'}}\quad\text{in }\mathbf{F}_p,$$

and both sides vanish if $Ш[p]\neq 0$.

This is a statement at one prime p. It relates no two primes, does not show that the common value is nonzero, and does not derive the value 288. At $p=5$ it only compares eligible pairs, and it says nothing at 19, 59 or 487. A mismatched generator changes the ratio by a square factor, which can be -1; a change of period scales every ratio by one common constant.

10. Stopped routes

We group the stopped routes by the kind of stop. Entries labeled Research record summarize archived arguments whose full definitions and proofs are in the claim-index evidence; those summaries are not self-contained exclusion theorems. Elementary counterexamples below are stated directly.

- A *proved obstruction* is a theorem about an explicitly delimited class of constructions. Nothing is claimed outside that class.
- An *equivalent reformulation* is a target proved equivalent to the open problem. It is not an impossibility.
- An *unaffordable* route is a budget decision.
- An *unsuccessful attempt* claims nothing.

10.1. Proved obstructions.

Obstruction 10.1 (`X.1`; elementary)**.** *Integral sources do not bound prime support. Two examples show this:*

- *The maps $\mathbf{Z}\to\mathbf{Z}_p$, $1\mapsto p^2$, are nonzero at every prime and have nonunit image at every prime.*
- *The two-term complex over $\prod_p\mathbf{Z}_p$ with differential $(p^2)_p$ is perfect, yet has finite nonzero defects at infinitely many primes.*

Hence source integrality, profinite perfectness, pointwise nonvanishing and pointwise finite primary groups do not, individually or together, imply finite support. By contrast, one fixed nonzero integer C with $Ш[p]\neq 0\Rightarrow p\mid C$ would give finite support.

Obstruction 10.2 (`X.2`; elementary)**.** *Suppose the localization matrix of $\langle P_1,P_2\rangle/p$ is invertible. That says nothing about the quotient $\mathrm{Sel}_p/(E(\mathbf{Q})/p)$: in an exact sequence $0\to M\to M\oplus Q\to Q\to 0$, an isomorphism on M carries no information about Q.*

Research record 10.3 (`X.5`; written proof, in the stated class)**.** Proposition 5.4 has no realization inside the finite logarithmic class L3. Four facts are established:

- Nonconstant polynomials in $\ell(z)$ have infinite cusp-translation span.

- The ordinary algebraic infinitesimal Tate deformation does not give the required jet.
- The half-ray maps to a null-homotopic loop in E.
- No finite algebraic connection, iterated integral or finite exact correction in the class L3 supplies the required local identity.

The half-ray and full-ray moments satisfy $J_j = (1 + (-1)^j)H_j$.

Research record 10.4 (`X.6`; computer-assisted)**.** A normalized, Atkin–Lehner-invariant second derivative $\mathcal{E}''(z, 0)$ of real-analytic Eisenstein series satisfies

$$\int_\gamma \alpha_f \, \mathcal{E}''(z, 0) = \tfrac{1}{2}L''(E, 1) + \Delta,$$

with an explicit Fourier residual Δ. Two single-vertex completions have certified residuals:

- the midpoint maximal-compact completion: $0.000535614030 \le \Delta_{\mathrm{mid}} \le 0.000801512867$;
- the Fricke-symmetrized endpoint completion: $0.000757821864 \le \Delta_* \le 0.000841639764$.

Both residuals are positive, so neither completion reproduces the weighted Mellin period. Arbitrary mixtures in the invariant span lie outside this class.

Research record 10.5 (`X.7`; written proofs, each in its stated scope)**.** Four further analyses of the midpoint residual each stop in their stated scope:

- The U_{4547}-trace forces the translated residual periods to sum to zero, but it does not force the zeroth period to vanish.
- The archived calculation excludes the specified finite combinations of ordinarily convergent Euler-product Dirichlet families and their derivatives by a Mellin-transform uniqueness argument with a nonzero continuous density. The precise representation and convergence class is part of that record; this summary excludes neither every conceivable Euler-product identity nor isolated special-value identities.
- The residual matches the Berndt–Kühn weight-two series after differentiation, but it has no arithmetic Green-current identification.
- The relative/holomorphic Deligne-regulator interpretation on compact $X_0(N)$ is obstructed: the curvature is nonzero and has no smooth extension across the cusp.

Obstruction 10.6 (`X.8`; written proof)**.** *Let J be a global C^1 scalar function on $Y_0(N)$ such that J and its first derivatives have polynomial-logarithmic growth at every cusp. Then for every holomorphic weight-two cusp form,*

$$\int_{Y_0(N)} \alpha_f \wedge dJ = 0.$$

Proof. $\alpha_f \wedge dJ = d(-J\alpha_f)$, and the cusp fluxes vanish by cusp decay. An independent unfolding of the Eisenstein family gives the same result. □

The class excludes singular, multivalued and nonexact kernels, and kernels with nonzero boundary flux.

Obstruction 10.7 (`X.10`; elementary)**.** *Let k be a number field, A/k an abelian variety, and $0 \to \mathbf{G}_m^s \to \mathcal{G} \to A \to 0$ a commutative extension. Then*

$$Ш^1(k, \mathcal{G}) \xrightarrow{\sim} Ш(k, A).$$

The same holds for vector-group extensions and for quasi-trivial tori. For tori split by an extension of degree prime to ℓ, the ℓ-primary parts agree. In particular, the extension of E by the split torus $\mathbf{G}_m^2$ attached to P_1, P_2 retains exactly the unknown group; it is also anti-affine [6].

Obstruction 10.8 (`X.11`; elementary)**.** *Totally positive quadratic integers can approach* 36 *while matching arbitrarily much prescribed finite local data. The first-derivative formulas tested in the records have vanishing inputs for this rank-two setting. The bare factorization* $L(E/K, s) = L(E, s)L(E^K, s)$ *supplies no arithmetic second-derivative comparison by itself. These are scoped tests, not exclusions of every future formula.*

Obstruction 10.9 (`X.13`; elementary)**.** *At a supersingular* $p \geq 5$*,* $a_p = 0$*. So* $X^2 + p$ *has no unit root, and ordinary stabilization cannot be used; one has instead* $\mathrm{pr}^2(\theta_{n+2}) = -p\,\theta_n$*. The signed objects of Theorem 7.3 replace the unit-root lane.*

The compression obstruction for the 2243 certificate (`X.15`) is Proposition 8.4.

10.2. **Failure of the proposed trace-divisibility inference.**

Obstruction 10.10 (`X.3`; counterexample)**.** *The unrestricted implication tested in the project is*

$$\alpha \in \mathfrak{P},\ \mathfrak{P} \mid p \implies \mathrm{Tr}(\alpha) \in p\mathbf{Z}.$$

This is false: $4 + i \in \mathbf{Z}[i]$ *lies above* 17 *but has trace* 8*. This elementary counterexample alone is not a source-level refutation under an unstated hypothesis set. The separate Destrempes–Malinin audit is retained in the claim index with its source-specific scope. No exceptional-prime set is imported from that preprint here.*

10.3. **Equivalent reformulations.**

Research record 10.11 (`X.9`; written deduction applying published theorems)**.** Varying-level visibility covers every relevant class for each fixed residual prime (Tamiozzo [39], with Zhang's hypotheses [42]). It gives no horizontal gain, for three reasons:

- An auxiliary prime ℓ serves p only if $p \mid D_\ell = (\ell + 1)^2 - a_\ell^2$, so a finite stock of auxiliary primes serves finitely many p.
- Functorial descent to a fixed ambient variety that preserves the class up to a p-unit is obstructed.
- For a fixed embedding, eventual fixed-ambient visibility on the ordinary tail is *equivalent* to eventual ordinary vanishing.

Research record 10.12 (`X.12`; elementary)**.** Over $\mathbf{Q}(\sqrt{-23})$, where $\mathrm{rank}\, E(K) = 3$, a fixed rank-three ambient base-change model within the finitely generated module class is equivalent to finite lane support. A bounded variant gives only $\#Ш[p] \leq p^{B-3}$.

Research records 9.1 and 9.2 are equivalences of the same kind.

10.4. **Unsuccessful attempts and procedural stops.**

- *Eisenstein congruences* (`X.4`). Nonclassical weight-two methods reach only the Selmer subspace of extensions that factor through a chosen automorphic quotient, and nothing bounds the complement. Separately, $1 \oplus \rho_E$ has Hodge–Tate weights $\{0, 0, 1\}$, which are irregular for classical GL(3) cohomology. The defect is one of coverage, not of arrow direction.
- *The prime-by-prime ladder* (`X.14`). It was stopped at 13, resumed through 47 with the singleton certificates of Theorem 4.15, and stopped again. Settling the smallest remaining prime moves only the conditional threshold (now 25281, with 53 next) and never reaches the infinite tail; a prime settled out of order does not move it. The depth-three requirement at 17 belonged to the cyclotomic criterion, not to the singleton route. No finite list of settled primes establishes finite support.
- *Other named constructions* (`X.17`). Each was stopped without an impossibility theorem:
  - the cross-cusp kernel RS2 was archived by a stopping rule;
  - for the Heegner–Stark ideals I_3, no method of evaluation was found;

– the anticyclotomic witness $c_2(p)$ remains open at every prime (it is a different object from the singleton coefficient $C_2(p,\ell)$ of §4.5);
– a Saito–Kurokawa tangent inequality points in the wrong direction for this use;
– the generalized Kato classes of the H9 program lack a comparison with the rational points;
– one Castella–Hsieh branch misses the calibrated line at $p = 5$.

10.5. **Unaffordable.** The earlier cost models estimate the certificates of Theorem 8.1 and Criterion 8.5 (`X.16`) at about $1.1 \cdot 10^5$ and $6.3 \cdot 10^5$ core-hours, and the 2243 evaluator's restart defect is to be fixed first. A second model, for Criterion 8.7 at the smallest auxiliaries ($\ell = 27283$, 35059 and 71777 for 4547, 5843 and 2243), gives about $2.4 \cdot 10^4$, $3.2 \cdot 10^4$ and $7.4 \cdot 10^4$ core-hours for coefficient production, verification and accumulation; merging, input and output, restarts and acceptance are not yet priced, and in the current evaluator the retained checkpoint files alone would need at least 0.8, 1.3 and 5.8 terabytes. These are model estimates, not necessary workloads and not lower bounds for other algorithms. All of these computations are held by decision. This is not a mathematical stop.

11. What remains, and the structure of the problem

11.1. **The obligations.**

Open obligation 11.1 (`O.1`, FIN)**.** $Ш(E/\mathbf{Q})$ is finite. Equivalently, every primary component is finite and $Ш[p] = 0$ for all but finitely many p.

Open obligation 11.2 (`O.2`, ORDER9)**.** If Ш is finite, then $\#Ш = 9$. Given FIN, it suffices to show $\#Ш < 2601$. The bound $\#Ш[n] < 2601$ for all n is equivalent to FIN *together with* ORDER9 (Proposition 4.13), not to this conditional assertion alone.

Open obligation 11.3 (`O.3`, Q36)**.** $q = 36$. No comparison between a p-adic leading term and the real regulator is available. (This is narrower than saying that p-adic methods cannot give q.) Even a proof of $q = 36$ would leave ORDER9.

Together these three obligations are equivalent to full BSD for E (Theorem 3.7): the group obligation is finiteness together with order nine, and the real obligation is $q = 36$. FIN does not imply ORDER9; given FIN, the settled components allow $\#Ш = 9t^2$ with every prime factor of t at least 53. Every other statement in this paper is settled at its stated grade (Appendix A), so these endpoint obligations form the current known frontier. The claim index also keeps `B.8` (the unevaluated witness at 4547, a regime within the prime-by-prime strategy) and `O.4` (Question 11.4, a proposed mechanism) open; neither is a further endpoint obligation. The frontier describes the endpoint; it is not a count of missing lemmas. In this revision:

- the rank, the basis and the primary components at $p \le 13$ (`R.1`, `R.4`, `S.2`, `S.3`, `S.5`–`S.13` and `S.P`) do not depend on an unrefereed preprint; the components at $17 \le p \le 47$ (`S.17-47`) apply Ota's theorem as stated in its arXiv version, and the conditional route results `B.9`, `B.13`, `B.17`, `B.18` and `B.19` keep their explicit preprint dependencies;
- the deductions of the primary components at $3, 5, 7, 11, 13$ are Lean-checked from their listed hypotheses (Remarks 4.4 and 4.10);
- the 2-primary component is an exact computation;
- finiteness of $Ш[3^\infty]$ and the lower bound $Ш[3] \supseteq (\mathbf{Z}/3)^2$ do not depend on the main conjecture (Theorems 4.6 and 4.7); nor, by route K (Theorem 4.8), does the upper bound $\#Ш[3^\infty] \le 9$, so the exact order 9 is proved without it; Skinner's theorem is kept as a second route (Remark 4.9).

Under the retained prime-by-prime strategy, FIN and ORDER9 need every good prime $p \ge 53$ and the bad primes 2243, 4547 and 5843 (the table below). Prime-by-prime work cannot finish on its own, and a proof need not use that strategy. For Q36, rationality of q with denominator at

most 174 would already give $q = 36$ (Proposition 5.1); bounded algebraicity (Proposition 5.2) and the common certificate of Proposition 5.3 are other sufficient routes. None of these antecedents is proved.

The primes not yet settled fall into five regimes:

regime	available sufficient criteria
good ordinary $p \geq 53$	the singleton criterion of §4.5 and Criterion 7.1, computable prime by prime; Theorem 7.5 and Corollary 7.6, not evaluable
good supersingular $p \geq 53$ $(131, 167, \ldots$; infinitely many [15])	the singleton criterion and Criterion 7.2, computable prime by prime; Theorem 7.3, not evaluable
2243	Theorem 8.1: one coefficient, uncomputed; Criterion 8.7, unevaluated
5843	Criterion 8.5: one coefficient, uncomputed; Criterion 8.7, unevaluated
4547	the retained two-prime Kurihara sufficient witness and Criterion 8.7, both unevaluated

Every criterion in the table works at a single prime. FIN would follow from vanishing outside a finite set, together with finiteness at each remaining prime. Separate theorems may handle ordinary, supersingular and bad primes; one theorem need not cover every regime. Finite prime support alone does not control the depths at the exceptional primes.

Targeted screens of the literature, with the screened sources recorded with the evidence, found no published theorem that supplies a uniform-in-p input of either kind for any elliptic curve of rank at least two: neither nondegeneracy of the p-adic height pairing for almost all p, nor nonvanishing modulo p of depth-two Mazur–Tate coefficients uniformly in p. Such inputs are known one pair (E, p) at a time, by rigorous computation, as in §4. This describes the current known frontier of the sources read; it is not a claim that no such theorem exists.

11.2. **The structure of the problem.** The following observations come from reading the record as a whole. Items marked *(builder)* are ours and are not in the cited records.

(a) *One shared import (builder).* Skinner's multiplicative theorem supports both multiplicative bad-prime routes (Theorem 8.1, Criterion 8.5) and the second route to the only nonzero primary component (Theorem 4.5), whose primary support is route K. An actual defect in the imported theorem would affect these chains together. The failure of the project's attempted internal reconstruction does not establish such a defect. The good-prime components rest on Kato's divisibility instead, which is independent of it.

(b) *One computational trust base (builder).* Every analytic certificate is an exponentially weighted twisted L-series evaluated from overlapping coefficient corpora sharing one verifier lineage. Independent replays share the raw coefficients. A review of the integer coefficient verifier is therefore the single most valuable audit on the computational side.

(c) *Two lanes with a similar unresolved requirement.* The Stark lane over $\mathbf{Q}(\sqrt{-23})$ and the definite-theta lane over $\mathbf{Q}(\sqrt{-7})$ both lack an arithmetic evaluation of their specified witness; no identification of those witnesses has been proved. Neither provides a prime-independent integral object whose reductions control the p-adic witness. In analytic rank one that object is the Heegner index of a conductor-one Heegner point. For this curve the corresponding index over $\mathbf{Q}(\sqrt{-23})$ is infinite, since the rank there is three.

Question 11.4 (`O.4`)**.** Is some sufficient witness for this curve the specialization of a single integral object that does not depend on p, and whose nonvanishing or defect is controlled by one fixed integer? Candidate witnesses include $M_2(p)$, $I_3(p)$, C_p and h_p. A positive answer that also covers the supersingular and bad primes would give finite prime support. Lemma 9.6 shows that, once such an integral model exists, the effective constant follows formally.

11.3. **Is this a good place to start?** The assessment in this subsection is ours and is not in the cited records.

In favour.

- The rank and analytic rank are certified.
- There is an explicit basis with a certified regulator.
- q is certified within 0.006 of 36.
- The components at $p \le 47$ are settled, and the only nonzero one, $(\mathbf{Z}/3)^2$, is explicit.
- Certified auxiliary quadratic fields exist in rank two and in rank three.
- A conditional criterion, with a Lean-checked core, covers every good prime $p \ge 5$ with $p \nmid 7N$.
- Many mechanisms have been tested, and their failure points are recorded with scope.

Against.

- The conductor $6.5 \cdot 10^{15}$ makes naive full modular-symbol or quaternionic spaces have scale 10^{14}–10^{16}.
- The conductor is not squarefree, which excludes several published theorems in their printed form.
- The additive, potentially supersingular prime 4547 has no executable certificate in the retained implementation.
- The split prime 2243 carries an exceptional zero.
- The exact order of the nonzero primary component rests on route K, a written derivation read by model readers but not by a human referee; Skinner's printed theorem gives a second route, with two steps not established for this curve (Remark 4.9).

We leave to experts the judgment whether this balance makes E a useful test case for rank-two methods. This paper and its repository are organized to make that judgment possible.

Appendix A. Grades, reviews and the claim index

The accompanying repository contains `claims/CLAIM_INDEX.md`, generated from one specification. Each entry gives the statement, its kind, its evidence grades, imports, dependencies, normalization, verification record and qualifications, together with the supporting artifacts and their SHA-256 hashes. The kinds are:

- fact, theorem, conditional, equivalence, obstruction, open;
- historical (results about other curves);
- withdrawn (assertions made and later retracted; listed so that they are not revived).

Review levels. No statement has had independent human review. Some statements were reviewed by automated reviewers that were language models. The index records whether the reviewer was of the same model family as the tools that produced the argument, or of a different family. Such reviews are not counted as independent proofs: agreement among several models is not equivalent to several independent proofs. Independence, where it exists, comes from three sources:

- checking statements at the source;
- replaying computations through different code paths;
- formal verification.

Lean. Lean-checked statements are deductions from explicit hypotheses. The hypotheses (arithmetic inputs imported from the literature, and structural facts) are listed, and are not themselves formalized unless the index says so.

Stops. Stopped routes are classified as proved obstructions, defective sources, equivalent reformulations, unaffordable routes, unsuccessful attempts or procedural stops (§10). No stop is used as evidence against a route outside its stated scope.

Sufficient witnesses. Nonvanishing of a coefficient such as C_p, $M_2(p)$ or K is a sufficient witness. Its failure refutes nothing, and the obligation it serves may be met by other means.

AI assistance. This project used Codex, Claude and Aristotle to propose and examine mathematical arguments, write and review code, perform computations, construct formal proofs, and assemble this manuscript under the author's direction. AI reviews are not human peer review. The named software checks verify only their stated computations or formal implications; they do not validate unformalized arithmetic imports automatically. The first release review and its scope are recorded separately.

Appendix B. Computations and their trust base

Coefficients. Prime coefficients a_q are produced by an untrusted fast program. Each is then checked by an independent integer verifier, either by exhaustive point counting for small q or by an integer group-order witness. The verified values are stored in segments covering intervals of 2^{24} integers (one coefficient per prime in the interval), which are hash-chained and bound by manifests; the composite coefficients a_n are generated multiplicatively. The corpus used here reaches $T = 5.5 \cdot 10^{10}$. It is not in the repository (`data/LARGE_DATA.md`) and is available by arrangement or by regeneration.

First-review repair. The legacy group-order verifier failed to require its listed prime factors to be distinct. A concrete false certificate at $q = 4099$ was accepted: $a_{4099} = 68$ instead of the exact value 86. The repaired verifier rejects it and guards the factor product against overflow. A separate read-only scan of all 3757 retained witness files checked distinct factors and safe product bounds on 2,693,070,671 records (including overlaps), and matched every file to its retained hash. All passed. This supplements the retained original group-arithmetic checks; it is not a fresh end-to-end coefficient replay. Original sources are preserved, with the repair and adversarial regression under `review/repairs/`. The mathematical acceptance predicate and preservation argument, including the original-receipt join, are given in Appendix F. This does not claim a new large group-arithmetic replay.

Interval evaluation. Symbols and L-values are evaluated with Arb [19]. Each series is truncated at a cutoff chosen from an explicit elementary tail bound, and the tail bound is added to the ball radius. A modular symbol is *identified* only when its ball has radius below $1/2048$ and contains exactly one point of $\frac{1}{1024}\mathbf{Z}$. Downstream arithmetic is exact, and each coefficient was computed twice by independent exact methods.

Trust base. The trust base for the numerical certificates consists of:

- the verifier and engine sources (included);
- FLINT/Arb 3.3.1, GMP, MPFR;
- the compiler, operating system and hardware.

No certificate has a second, independent end-to-end implementation, except for small-scale cross-checks. The regulator certificate uses directed-rounding decimal arithmetic in Python, and the 2243-adic height certificate uses only the Python standard library.

Formal verification. The Lean development for Theorem 7.3 builds with Lean 4.32.2 and Mathlib at tag v4.32.2. The build uses only the standard axioms. Replay instructions are in the repository.

Exact replays in the repository. The script `tools/verify_release.py` checks every evidence hash and reruns the following:

- `tools/replay/curve_invariants.py`, which recomputes the invariants, the minimal model, the points and the supersingular primes below 2000;
- `tools/replay/coefficients_from_symbols.py`, which recomputes every certified coefficient from the certified symbol tables;
- the 2243-adic height certificate and its separate verifier;
- the exact rank computation of Proposition 8.4;
- the Atkin–Lehner pairing on the certified tables at 27 and 81;
- optionally, the Lean core.

All pass; the run is recorded in `review/VERIFICATION_RUN.md`. The large certificates cannot be replayed without the coefficient corpus.

Appendix C. Contradictions and withdrawn claims

`claims/CONTRADICTIONS.md` lists every discrepancy found while this package was assembled. Resolved items are kept visible. Six are unresolved:

(1) Stein–Wuthrich's multiplicative "if and only if" and its attribution to Jones. This is a literature question, and it is not used here.
(2) A disputed exceptional zero at 5843 for the twist E^{-7}. It is off the critical path.
(3) Three changed metadata files in the external C_{13} record. Appendix F.3 records their mismatches and the intact calculation-input chain; the historical status remains unrecovered.
(4) The canonical master is intentionally frozen. Later results supersede it additively; this release does not edit that master.
(5) Two reviewers grade the 2243 restart defect differently.
(6) The canonical proof of Theorem 7.5 has not been chosen between the two routes.

The withdrawn register (`CLAIM_INDEX.md`, ids beginning with `W`) includes:

- a claimed unit specialization at 13;
- an "arrow direction" admission rule;
- superseded values of C_{13};
- overstatements of the order of the theta element and of the regulator input;
- corrected coefficient relations;
- superseded cost figures;
- the evaluator's isolation claim.

Older withdrawals are in the master record's frozen register.

Appendix D. The numerical Manin bound and the symbol lattice

This appendix supplies the composition omitted from the earlier manuscript. It transcribes the governing, corrected proof in `evidence/src/BSD_MANIN_2_PRIMARY_GATE_2243_12_2026-09-18/00_REPORT.md`, §§1–3 and 6. It is a written proof using the geometric inputs identified below, not a formal verification of integral modular-curve geometry. The later source audit and its corrections are in `review/SECOND_REVIEW.md`. In particular the integer eight is derived here, not attributed verbatim to Pasten or CNS.

D.1. **The chosen parametrization.** Let $\pi : J_0(N) \twoheadrightarrow E_0$ be the optimal quotient for f, and let $\phi_0 = \pi \circ \iota_\infty$, with $\iota_\infty(x) = [x - \infty]$. Use minimal Néron differentials and choose signs so that $\phi_0^* \omega_0 = c_0 \omega_f$, where $\omega_f = 2\pi i f(z)\, dz$. The retained isogeny-class computation identifies two curves, E and its rational 2-isogenous partner. Thus the map used for E is $\phi_E = \phi_0$ if $E = E_0$, and otherwise $\phi_E = \psi \circ \phi_0$, where $\psi : E_0 \to E$ is that 2-isogeny. This definition does not allow

subsequent composition with an arbitrary multiplication map. For $\psi^*\omega_E = c_\psi\omega_0$, integrality and the dual isogeny give $c_\psi \mid 2$, hence $c_E \mid 2c_0$; see [11, proof of Cor. 7.3].

The odd-prime support theorem [1, Cor. 3.7] restricts the odd divisors of c_0 to primes whose squares divide N, here only 4547. Both displayed isogeny-class members have type III at 4547, with $(v(c_4), v(c_6), v(\Delta)) = (1, 2, 3)$ and $j \equiv 1728$. Since $4547 \equiv 3 \pmod 4$, this is potentially supersingular reduction. Theorem 4.6.3 of [14] excludes 4547 from c_0: its hypotheses $p > 7$, the stated additive-type exclusions, and absence of a rational p-isogeny all hold (the last follows also from Theorem 2.3). Therefore c_0 is a power of two. The direct thesis check concerns that theorem's statement, not a new proof of the thesis.

D.2. **Reduction to an integral differential lattice.** Put $R = \mathbf{Z}_{(2)}$, $X = X_0(N)_R$, and

$$S = S_2(\Gamma_0(N), R), \qquad H = H^0(X, \omega_{X/R}), \qquad e' = \min\{e : 2^e S \subseteq H\}.$$

Here S denotes the forms with integral expansion at ∞. Because $3 \mid N$ and $3 \equiv 3 \pmod 4$, [11, Thm. 6.12(c), Prop. 6.2] identify H with the Néron differential lattice of $J_0(N)$. Their Theorem 5.15 puts ω_f in H. Its first coefficient is one, so $H \cap \mathbf{Q}\omega_f = S \cap \mathbf{Q}\omega_f = R\omega_f$.

Let P_f be the Petersson projection onto $\mathbf{Q}\omega_f$, and choose a generator η of $P_f(H)$ with $\omega_f = 2^\kappa\eta$, after absorbing an R-unit. The prime-to-N Hecke operators are self-adjoint, and strong multiplicity one makes their common f-eigenspace one-dimensional. Dualizing $\pi t = \chi_f(t)\pi$ therefore shows that $(\pi^\vee)^*$ factors through P_f. Writing $(\pi^\vee)^*\eta = u\omega_0$ with $u \in R$, the identity $\pi\pi^\vee = [\deg \phi_0]$ gives

$$v_2(\deg \phi_0) \geq v_2(c_0) + \kappa.$$

The modular-exponent divisibility [2, Thm. 3.6(a)] gives the opposite bound

$$v_2(\deg \phi_0) \leq v_2(P_f(S)/R\omega_f) \leq e' + \kappa.$$

The last inequality follows by projecting $2^{e'}S \subseteq H$. Cancellation yields $v_2(c_0) \leq e'$. This is the integral-lattice part of Pasten's method [33, §10]; it does not invoke an unspecified numerical constant from that paper.

D.3. **The explicit estimate $e' \leq 8$.** Use the regular rigidified modular model X_m for $\Gamma_0(N) \cap \Gamma_1(5)$ and its forgetful map $\alpha : X_m \to X$. Put $X' = X_{\Gamma_0(N/8)\cap\Gamma_1(5)}$, and let μ and c' be its index in $\mathrm{PSL}_2(\mathbf{Z})$ and cusp count. The imported geometric facts are:

(i) X' has good reduction at two and $2g' - 2 = \mu/6 - c'$. Its geometric supersingular locus has $s = \mu/12$ points.
(ii) The special fibre of X_m has four geometrically integral components C_i, indexed by $(3, 0), (2, 1), (1, 2), (0, 3)$, each of multiplicity one. Their arithmetic genera are at least g'.
(iii) Each pair meets over every supersingular point, so $C_iC_j \geq s$ for $i \neq j$. The full fibre $F = \sum_i C_i$ satisfies $FC_i = 0$.
(iv) The generic fibre has index 12μ and $4c'$ cusps, whence $\omega_{X_m/R}F = 2\mu - 4c'$. The map α is *generically* ètale on each special-fibre component.

Regularity, component multiplicities and generic ètaleness are taken from the Katz–Mazur inputs as used in [11, §5.9, Lemmas 5.8, 5.10]; the supersingular subgroup is $\ker F^3$ in their proof of 6.10. The genus and mass formulas and adjunction are standard imports. The original Katz–Mazur and adjunction proofs were not independently replayed in this release. This precisely identifies the geometric portion requiring expert review.

For $s_0 \in S$, write the divisor of $\alpha^* s_0$ as $D_{\mathrm{hor}} - \sum_i k_i C_i$, where D_{hor} is effective and horizontal. Integral expansion at ∞ gives $k_\infty \leq 0$; other k_i may also be negative. If $K = \max_i k_i > 0$, choose

j attaining it. Since $FC_j = 0$,

$$\omega C_j = D_{\mathrm{hor}} C_j + \sum_{i \neq j} (K - k_i) C_i C_j \geq K C_\infty C_j \geq K\mu/12.$$

This formula includes any positive vertical zero at C_∞ and does not silently set it to zero. Adjunction gives, for every other component,

$$\omega C_i = 2p_a(C_i) - 2 - C_i^2 \geq \mu/6 - c' + 3\mu/12.$$

Subtracting the other three contributions from ωF yields

$$K\mu/12 \leq 2\mu - 4c' - 3(\mu/6 - c' + \mu/4) = 3\mu/4 - c', \qquad K \leq 9 - 12c'/\mu < 9.$$

Thus $K \leq 8$; if $K \leq 0$ there is nothing to prove. Multiplying by 2^8 removes every vertical pole upstairs. Generic ètaleness transfers the codimension-one regularity downstairs, and normality together with the reflexive dualizing sheaf extends across codimension two. Hence $2^8 S \subseteq H$ and $v_2(c_0) \leq 8$.

D.4. **From the bound to the exact lattice.** It follows that $c_0 \mid 2^8$, $c_E \mid 2^9$. A cusp $r = a/d$ with $\gcd(d, N)$ an exact divisor of N is rational over $\mathbf{Q}$. Manin–Drinfeld makes $\phi_E(r) - \phi_E(\infty)$ rational torsion, of exponent dividing two. Consequently $2c_E\lambda(r)$ belongs to the Néron period lattice of E. Because $\Delta > 0$, that lattice is rectangular, with least positive periods Ω^+ and $i\Omega^-$. Taking the two coordinates proves

$$[r]^{\pm} \in (2c_E)^{-1}\mathbf{Z} \subseteq 2^{-10}\mathbf{Z}.$$

This applies separately to denominators $27, 81, 5, 25, 7, 49, 11, 121, 13, 169$ and the smaller retained levels; it is not a claim about all cusps. There is *no nine-power spare margin*: the worst-case denominator here is exactly 2^{10}.

Appendix E. The signed bridge: proof and formalization boundary

This is the written deduction behind Theorem 7.3, supplied for review rather than replaced by a build-success label. The complete retained argument is `evidence/src/BSD_SUPERSINGULAR_SIGNED_BRIDGE_2026-09-22/SIGNED_MINIMAL_LAMBDA_BRIDGE.md`. The source review and Lean report are separate evidence; no theorem of BLV is formalized here.

E.1. **Arithmetic inputs and conventions.** Let $\Lambda = \mathbf{Z}_p[[T]]$, $\gamma = 1 + T$, $\iota(\gamma) = \gamma^{-1}$, and let X be the Pontryagin dual of the minus signed Selmer group over the anticyclotomic extension of $K = \mathbf{Q}(\sqrt{-7})$, with the ι-twist required by the sesquilinear Poitou–Tate pairing. Replacing X by X^ι preserves all augmentation quotients and conjugation eigenspace ranks used below. Write $F = F^-$. The following are the arithmetic imports and source-comparison obligations of the argument:

(I1) *Control:* $X/TX \simeq \mathrm{Sel}_{p^\infty}(E/K)^\vee$, compatibly with complex conjugation τ. The local and control inputs are [5, Props. 5.3, 5.9, Cor. 5.5], with the retained bad-place comparisons.

(I2) *Characteristic and exact specialization:* $\mathrm{char}_\Lambda X = (FF^\iota)$ by [5, Thm. 7.3]. At a character $\gamma \mapsto 1 + a$, $a \in p\mathbf{Z}_p$, where both factors are nonzero,

$$\mathrm{length}_{\mathbf{Z}_p} X/(T - a)X = v_p\big(F(a)F((1 + a)^{-1} - 1)\big).$$

This is the exact, non-exceptional minus case of [5, Thm. 7.1], with Proposition 5.9 and Remark 5.16. One applies the finite-coefficient theorem for arbitrarily large depths; the asserted bounded lengths stabilize and give the full specialization. Equality, not merely characteristic-ideal divisibility, is essential.

(I3) *Equivariant presentation:* over $A = \mathbf{F}_p[[u]]$, where $u = \gamma - \gamma^{-1}$, the residual Poitou–Tate map gives

$$F_1 \longrightarrow F_0 \longrightarrow W = X/pX \longrightarrow 0,$$

with F_1, F_0 free of equal rank and τ-equivariant. Here F_1 is the relaxed compact signed Selmer module for a freeing set and F_0 the sum of its singular local groups. The freeing primes are inert in K, hence stable under τ; see [5, §3.2, Lemma 3.1, Prop. 5.13]. Finite-level Poitou–Tate followed by inverse limits of finite groups gives this presentation; Corollary 5.14 identifies the relaxed limit. No integral Selmer-complex freeness is postulated.

At $u = 0$, Proposition 5.10's injection is an isomorphism by the rank calculation in Proposition 5.13 at the base and infinite levels. The specialized kernel is the classical compact p-Selmer group and the cokernel its discrete dual. Naturality of the Weil pairing and local invariant maps under τ identifies their individual plus and minus dimensions. This equivariant identification is a written arithmetic comparison in the retained proof, not a consequence of equality of their total dimensions and not a theorem checked by Lean.

(I4) *Rank and descent:* for odd p, restriction/corestriction identifies the plus part of the classical Selmer dual over K with that over $\mathbf{Q}$. Kummer theory and Theorem 3.1 give

$$0 \to Ш[p^\infty]^\vee \to \mathrm{Sel}_{p^\infty}(E/\mathbf{Q})^\vee \to \mathbf{Z}_p^2 \to 0.$$

The retained source review checks the cited BLV version and these comparisons. BLV's imported main-conjecture results remain published/preprint imports at their recorded grades; a successful algebra replay does not reprove them.

Lemma E.1 (Removal of the finite error)**.** *Assume (I2) and $\overline{F} \neq 0$. If $d = \mathrm{ord}_T \overline{F}$, then X is free over $\mathbf{Z}_p$ of rank $2d$.*

Proof. Weierstrass preparation gives $\mu(F) = 0$, $\lambda(F) = d$, so X is finite over $\mathbf{Z}_p$ and $Y = X/H$ is free of rank $2d$, where H is its finite $\mathbf{Z}_p$-torsion. Choose a nonzero $a \in p\mathbf{Z}_p$ away from the finitely many zeros of FF^ι. On Y, $T-a$ has nonzero determinant and is injective. The determinant-index formula gives

$$\mathrm{length}\, Y/(T-a)Y = v_p((FF^\iota)(a)) = \mathrm{length}\, X/(T-a)X.$$

The Tor term is $Y[T-a] = 0$, so specialization of $0 \to H \to X \to Y \to 0$ is exact on the left as well. Additivity of length gives $H/(T-a)H = 0$. Since $T-a$ is in the maximal ideal of Λ, Nakayama gives $H = 0$. □

Lemma E.2 (Conjugation rank balance)**.** *Under (I3) and Lemma E.1,* $\mathrm{rank}_{\mathbf{Z}_p} X^+ = \mathrm{rank}_{\mathbf{Z}_p} X^- = d$.

Proof. The parameter $u = T(2+T)/(1+T)$ generates the augmentation ideal, and $\tau(u) = -u$. The finite-dimensional space $W = X/pX$ has dimension $2d$ and nilpotent u-action U, with $\tau U = -U\tau$. Since the cokernel in (I3) is A-torsion and the free modules have equal rank, $F_1 \to F_0$ is injective. Reducing this resolution modulo u, the equality of the individual eigenspace dimensions of its kernel and cokernel gives

$$[F_0/uF_0] - [F_1/uF_1] = 0$$

in the representation group of $\langle\tau\rangle$. A second equivariant free resolution is

$$0 \to A \otimes_{\mathbf{F}_p} (W \otimes \mathrm{sgn}) \xrightarrow{u-U} A \otimes_{\mathbf{F}_p} W \to W \to 0.$$

Nilpotence of U proves exactness. The sign on the source is necessary for equivariance because both u and U reverse sign. The two resolutions compute the same equivariant Tor groups at $u = 0$,

so $[W]-[W\otimes\mathrm{sgn}]=0$. As p is odd, this says $\dim W^+=\dim W^-=d$. The idempotents $(1\pm\tau)/2$ split the free $\mathbf{Z}_p$-module X and commute with reduction modulo p, proving the assertion. □

Lemma E.3 (The rank-two quotient)**.** *Under (I1)–(I4), $d\geq 2$ and $d=2$ implies $Ш[p^\infty]=0$.*

Proof. The composite $X^+\twoheadrightarrow \mathrm{Sel}_{p^\infty}(E/\mathbf{Q})^\vee\twoheadrightarrow \mathbf{Z}_p^2$ is surjective, hence $d\geq 2$. At $d=2$ it is a surjective map of free modules of the same rank, so its determinant is a unit and its kernel is zero. Every element of the kernel of the second map lifts through the first and would lie in this zero kernel. Thus $Ш[p^\infty]^\vee=0$, and Pontryagin duality gives the result. This proves nothing about the minus eigenspace or the twist's Sha. □

If $\overline{F}=0$, its constant and linear coefficients vanish directly. Otherwise $d\geq 2$ gives the same conclusion. Let G_1 be the first-layer Galois group, let γ_1 be the image of γ in G_1, and put $t=\gamma_1-1$. Under the natural map $\Lambda\to\mathbf{F}_p[G_1]/(t^3)\simeq\mathbf{F}_p[t]/(t^3)$, the variable T maps to t. Writing $\overline{F}(t)$ for the image of F in this quotient, BLV's first-layer minus formula is $\overline{F}(t)=\sum_{j<p}v_j(1+t)^j$, where $v_j=\bar{\psi}(\gamma^jQ_1^{\mathrm{Gr}})$ and Q_1^{Gr} is the traced conductor-p^2 Gross point. Thus $\bar{b}_0=M_0$, $\bar{b}_1=M_1$, and $\bar{b}_2=(M_2-M_1)/2=M_2/2$. Nonzero M_2 gives $d=2$ and completes Theorem 7.3. This argument supplies no nonzero packet value.

E.2. **The precise Lean boundary.** The complete source signature, including typeclass and map hypotheses, is reproduced verbatim in `review/proofs/A3_SIGNATURE_AND_INPUTS.md`; the underlying file is `evidence/src/BSD_A3_LEAN_CORE_2026-09-22/A3Core.lean`. The correspondence is:

Lean hypothesis	arithmetic or structural justification
`c, hc, hcσ, ρ`	(I1), plus-part descent in (I4)
`hexact`	exact specialization in (I2)
`iA, iB, πA, πB, u1, u0`, exactness and commutation	the two graded pieces of (I3)
`harith`	equality of the minus kernel and cokernel dimensions in (I3)
`κ, hκ`	Kummer surjection in (I4); identifying its kernel with the Sha dual is external
`Module.Finite, hrank, hlam`	$\mu=0$, characteristic ideal, and $d=\mathrm{ord}_T\,\overline{F}$
`hnil, hinj, hchar`	topological nilpotence and determinant-index formula for Y
`hτ, hτu`	conjugation is an involution and $\tau u=-u\tau$

Exactness of eigenspaces (averaging by $(1\pm\tau)/2$), their compatibility with reduction, and Pontryagin duality are mathematical modeling steps. The sign-twist identity is formalized, but not every identification from Selmer theory to these abstract modules is. The prior pinned Lean replay is retained; unchanged Lean code was not rebuilt merely to revise this exposition.

Appendix F. The repaired coefficient predicate and custody

F.1. **Why a valid witness determines the coefficient.** For a good prime $q\geq 4096$, the binary record is $(q,a,x_0,k,(\ell_i,e_i)_{i=1}^k)$: an unsigned 64-bit prime, signed 32-bit proposed trace, two one-byte fields, and k pairs of an unsigned 64-bit factor and a one-byte exponent. Each file begins with its interval endpoints and record count. The verifier generates the primes in that interval independently. Small and bad primes are counted directly, not supplied by these records.

Put $A = -27c_4$, $B = -54c_6$, $d = x_0^3 + Ax_0 + B \neq 0$ in $\mathbf{F}_q$. The short curve with coefficients A, B is isomorphic to the seed over $\mathbf{F}_q$ for $q > 3$. On its quadratic twist

$$E_d : Y^2 = X^3 + Ad^2X + Bd^3$$

the point $P = (dx_0, d^2)$ is rational. With $\chi = (d/q)$ and $h = q + 1 - \chi a$, acceptance checks $|a| \leq 2\sqrt{q}$, $[h]P = O$, and, for every factor, primality of ℓ_i, $e_i \geq 1$, $e_i \leq v_{\ell_i}(h)$, and

$$[h/\ell_i^{v_{\ell_i}(h)-e_i+1}]P \neq O.$$

If n is the order of P, then $n \mid h$ and the last test forces $v_{\ell_i}(n) \geq e_i$. With *distinct* ℓ_i, the product $D = \prod_i \ell_i^{e_i}$ divides n, hence divides $\#E_d(\mathbf{F}_q)$. Set $s = \lfloor \sqrt{4q} \rfloor$. The checks $D > 2s$ and exactly one multiple of D in $[q + 1 - s, q + 1 + s]$, together with $D \mid h$, force $\#E_d(\mathbf{F}_q) = h$ by Hasse. Quadratic twisting then gives $a_q(E) = a$.

The distinctness condition is indispensable: repeating a factor does not increase the proved divisibility of n. The retained adversarial example at $q = 4099$ uses an order-six point and repeats the factors to manufacture $D = 576$, accepting the false trace 68; exhaustive counting gives 86. The repaired code rejects that example and accepts the separately constructed genuine control.

F.2. **Compositional repair, with its limits.** The exact patch is `review/repairs/wverify_distinct.patch`. It makes only two changes to the arithmetic predicate: it rejects repeated factors and replaces the product update by the guarded test $D \leq h/\ell_i$ before multiplying. On well-formed records in the retained range $q \leq 55046045695$, the following implication holds:

$$\text{old group/primality checks pass} \quad + \quad \text{distinct factors and safe products}$$
$$\Longrightarrow \quad \text{repaired predicate passes.}$$

Indeed, the prime-power tests and distinctness give $D \mid h$ over the integers, so every partial product is at most h and passes the new guard. The existing group operations and other acceptance conditions are unchanged. This is a restricted preservation argument, not an equivalence claimed for malformed inputs or executions with undefined overflow.

The supplemental scanner parses each full record, checks factor distinctness and each product in 128-bit arithmetic against the Hasse upper bound, checks strictly increasing primes within the declared interval, and rejects truncation, count errors and trailing bytes. Its bound keeps each intermediate product within the retained 64-bit range. It deliberately does not repeat primality or elliptic-curve operations.

All 3757 scanned files, comprising 2,693,070,671 records including overlaps, match the retained witness hashes. For each file the follow-up review also joined the scan to the original acceptance receipt by segment, interval and witness count. The full receipt texts, their hashes and the join are in `review/followup_logs/corpus_acceptance_receipts.json`; the portable receipt check verifies that composition. The six source runs contain respectively $9, 885, 467, 43, 1405, 948$ witness files. The overlapping prefixes are intentional: later runs reuse earlier verified coefficient segments. The C7, C11 and C13 binding files record that reuse separately.

What is retained as trust: the original execution receipts, their recorded association with the input bytes, the unchanged group/primality implementation, and the compiler/hardware. A hash is an identity check, not a proof that an execution happened. The new scan and receipt join close the specifically identified missing conditions at this declared trust grade; they do not constitute a fresh independent replay of billions of group calculations or a proof that the C program has no other defects. The repaired verifier is provided for future replay. The original 2243 pilot still contains the legacy source and is not thereby approved for execution.

F.3. **The C13 discrepancy, stated precisely.** Three files in the external C13 execution folder differ from their pre-change bindings: `RESULT.json`, `STATE.json`, and `RESUME.md`. They are status/summary records, not the inputs consumed by the arithmetic recomputation. The original status value in `STATE.json` has not been recovered, so the historical claim that these files were preserved unedited is not accepted.

The separate master completion record binds the denominator-13 and denominator-169 symbol tables and Arb outputs. All four match those bindings now. The bound input manifest, Stage C output, downstream script, downstream result and incident record also match. The completion binding itself matches the hash in the completion review. The exact old and current hashes of the three changed files and the nine intact calculation inputs are recorded in `review/followup_logs/c13_custody_reconciliation.json`. No old manifest or source file was overwritten to obtain these matches.

The portable downstream replay reads the intact symbol tables and returns $C_{13} = 2$ in the ω_1 normalization and 1 in the full-period normalization. Thus the changed status files are not arithmetic premises of that replay. This retains the 13-primary theorem at its stated computer-assisted grade, including the symbol-lattice theorem and shared coefficient/Arb trust base. It does not certify the entire execution history as unchanged, and it does not independently reevaluate the original large numerical series.

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